

$\Rightarrow$ Un-conditional Pdf = conditional Pdf
(or the other words)

$\Rightarrow$ Joint Pdf = product of two marginal Pdf

$$= \int_{-\infty}^{\infty} f_{2|1}(x_2|x_1) f_1(x_1) dx_1$$

$$= f_{2|1}(x_2|x_1) \int_{-\infty}^{\infty} f_1(x_1) dx_1$$

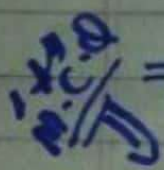
$\therefore$ Area under the curve
 $\therefore \int_{-\infty}^{\infty} f_1(x_1) dx_1 = 1$

$$f_2(x_2) = f_{2|1}(x_2|x_1)$$

Hence

$$f(x_1, x_2) = f_1(x_1) f_2(x_2)$$

(when $(f_{2|1}(x_2|x_1))$ does not depend upon x_1)



$\Rightarrow$ Proofs $\Rightarrow$ probability mass functions

$\Rightarrow$ X_1 and X_2 are independent to each other

If and only if

M-C

$$\Rightarrow P(x_1, x_2) = P_1(x_1) P_2(x_2)$$

5 marks 2022

Example:

Let the Joint Pdf of X_1 and X_2 be

$$f(x_1, x_2) = \begin{cases} x_1 + x_2 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Show That X_1 and X_2 are dependent.

So!

$$\text{let } f_1(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_2$$

$$= \int_0^1 (x_1 + x_2) dx_2 = \left[x_1 x_2 + \frac{x_2^2}{2} \right]_0^1$$

$$= (x_1(1) - x_1(0) + \frac{1}{2} - 0)$$

$$= x_1 + \frac{1}{2}$$

$$f_1(x_1) = x_1 + \frac{1}{2} \quad 0 < x_1 < 1$$

and

$$\Rightarrow f_2(x_2) = \int_{-\infty}^{\infty} f_1(x_1, x_2) dx_1$$

$$= \int_0^1 (x_1 + x_2) dx_1 = \left[\frac{x_1^2}{2} + x_1 x_2 \right]_0^1$$

$$= \frac{1}{2} - 0 + x_1 x_2 \Big|_{x_1=0}^{x_1=1} = \frac{1}{2} + x_2$$

$$f_2(x_2) = \frac{1}{2} + x_2$$

Now

$$\Rightarrow f_1(x_1) \cdot f_2(x_2) = \left(x_1 + \frac{1}{2}\right) \left(x_2 + \frac{1}{2}\right)$$

$$= x_1 x_2 + \frac{x_1}{2} + \frac{x_2}{2} + \frac{1}{4}$$

$$= x_1 x_2 + \frac{1}{2} x_1 + \frac{1}{2} x_2 + \frac{1}{4}$$

$$\text{put } x_2 = x_1$$

$$f_1(x_1) \cdot f_2(x_2) = x_1^2 + \frac{1}{2} x_1 + \frac{1}{2} x_2 + \frac{1}{4}$$

where

$$f(x_1, x_2) = x_1 + x_2$$

Since

$$\Rightarrow f(x_1, x_2) \neq f_1(x_1) \cdot f_2(x_2)$$

It is not independent

Topic # 112

Theorem: If X and Y are random variables with Joint distribution function F and marginal distribution function F_x and F_y , then X and Y are independent if and only if

$$F(x, y) = F_x(x) F_y(y)$$

For all $(x, y) \in \mathbb{R}^2$

By fundamental theorem of Probability

Proof

If X and Y are independent, then

$$\begin{aligned} P(x, y) &= P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y) \\ &= F_x(x) F_y(y) \end{aligned}$$

$\Rightarrow$ Now Suppose

$$F(x, y) = F_x(x) F_y(y) \text{ for all } (x, y) \in \frac{a^2}{b}$$

and let $A = (a, b]$ and $B = (c, d]$

$\Rightarrow$ Then

fundamental property

$$P(X \in A, Y \in B) = P(a < X \leq b, c < Y \leq d)$$

$F = F_x F_y$

$$= F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

$$= F_x(b) F_y(d) - F_x(a) F_y(d) - F_x(b) F_y(c) + F_x(a) F_y(c)$$

$\Rightarrow$ By using fundamental property of cdf

$$P(X \in A, Y \in B) = (F_x(b) - F_x(a)) (F_y(d) - F_y(c))$$

$$= P(a < X \leq b) P(c < Y \leq d)$$

$$= P(X \in A) P(Y \in B)$$

Hence

Thus X and Y are independent

Topic # 113

Theorem: Suppose that X_1 and X_2 are independent and $E(u(x_1))$, and $E(v(x_2))$ exist. Then,

$$E[u(x_1)v(x_2)] = E[u(x_1)]E[v(x_2)]$$

Proof

Let the independence of x_1 and x_2 are $f_1(x_1)$ and $f_2(x_2)$ by definition

$$E[u(x_1)v(x_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(x_1)v(x_2) f(x_1, x_2) dx_1 dx_2$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(x_1)v(x_2) f_1(x_1) f_2(x_2) dx_1 dx_2$$

$$= \int_{-\infty}^{\infty} v(x_2) f_2(x_2) \int_{-\infty}^{\infty} u(x_1) f_1(x_1) dx_1 dx_2$$

$$= \left[\int_{-\infty}^{\infty} u(x_1) f_1(x_1) dx_1 \right] \left[\int_{-\infty}^{\infty} v(x_2) f_2(x_2) dx_2 \right]$$

$$= E[u(x_1)] E[v(x_2)] \Rightarrow E[u(x_1)] E[v(x_2)]$$

Hence proved

For independent random variable
 $E(X_1, X_2) = E(X_1) E(X_2)$

m.c.Q

Topic # 115

Discrete Uniform Distribution

The discrete uniform distribution is a symmetric probability distribution whereby a finite number of values are equally likely to be observed.

M.C.Q

discrete uniform distribution

imp (with probability are equal)

rows (1) x column 2 $p(x) \Rightarrow$ probability

deno (can say) $= 1$

Example: let X denote the number of heads that we can have if a fair coin is tossed once. Then, we have

discrete uniform distribution

X	$P(X)$
0	$\frac{1}{2}$
1	$\frac{1}{2}$

Example: let X denote the number that we observe on the uppermost face if we roll a fair die once.

Then, we have

M.C.Q

X	$P(X)$
1	$\frac{1}{6}$
2	$\frac{1}{6}$
3	$\frac{1}{6}$
4	$\frac{1}{6}$
5	$\frac{1}{6}$
6	$\frac{1}{6}$

$\frac{6-1}{6}$

M.C.Q

Topic # 116

P = probability
 n = max
 P = function

PMF of Discrete Uniform Distribution

If X is assuming values $1, 2, 3, \dots, n$, then the probability mass function (PMF) of discrete uniform distribution is given by

$$P(X) = \begin{cases} \frac{1}{n}, & X = 1, 2, \dots, n \\ 0 & \text{elsewhere} \end{cases}$$

If $n=6 \Rightarrow P(X) = \begin{cases} \frac{1}{6} & X = 1, 2, 3, \dots, 6 \\ 0 & \text{elsewhere} \end{cases}$

n distinct equally-spaced values from 'a' to 'b' (at an interval of 1).

Then
$$P(X=x) = \begin{cases} \frac{1}{n} & x=a, a+1, a+2, \dots, b-1, b \\ 0 & \text{elsewhere} \end{cases}$$

where $n = b - a + 1$ M.C.Q

Example:

$$P(X=x) = \begin{cases} \frac{1}{b-a+1} & x=a, a+1, a+2, \dots, b-1, b \\ 0 & \text{elsewhere} \end{cases}$$

Here $a=1, b=6$ cdf M.C.Q
(cumulative distribution function)

Sol let

$$P(X=x) = \frac{1}{b-a+1}, \quad x=a, a+1, a+2, \dots, b-1, b$$

$$= \frac{1}{6-1+1}, \quad x=1, 1+1, 1+2, \dots, 6-1, 6$$

$$= \frac{1}{6}, \quad x=1, 2, 3, \dots, 5, 6$$

M.C.Q
cont. class
 all Tall, 1
 number of Tall 3
 zero in 6
 3/6 of head
 no. of head
 as 3/6
 as 1/2
(0,1)

Topic # 117

imp CDF of the Discrete Uniform Distribution

Example: Suppose That we are rolling one fair die one time and we want to have the cdf of the distribution in the mathematical form, algebraic form,

Sol. Step 1 - find the cumulative probabilities

X	p(x)	cumulative
1	1/6	1/6
2	1/6	2/6
3	1/6	3/6
4	1/6	4/6
5	1/6	5/6
6	1/6	6/6

Step-2

$P(x) =$	0	$-\infty < x < 1$
	$\frac{1}{6}$	$1 \leq x < 2$
	$\frac{2}{6}$	$2 \leq x < 3$
	$\frac{3}{6}$	$3 \leq x < 4$
	$\frac{4}{6}$	$4 \leq x < 5$
	$\frac{5}{6}$	$5 \leq x < 6$
	$\frac{6}{6}$	$6 \leq x < \infty$

The entire x -area (from $-\infty$ to ∞) must be covered

Topic # 118

Derivation of the Mean & Variance of the Discrete Uniform Distribution

$\Rightarrow$ **Mean:** we derive the formula for the mean of the discrete random variable $X \sim U(1, n) \Rightarrow (x=1, 2, \dots, n)$ each with probability $\frac{1}{n}$

$\Rightarrow$ **Formula:**

$$\mu = \sum_{x=1}^n x \cdot \frac{1}{n} = \frac{1}{n} \sum_{x=1}^n x$$

$$\mu = \frac{1}{n} (1 + 2 + \dots + n) = \frac{1}{n} \times \frac{n(n+1)}{2}$$

$$\mu = \frac{1}{n} \times \frac{n(n+1)}{2} = \frac{(n+1)}{2}$$

$\mu = \frac{n+1}{2}$

$\Rightarrow$ **Variance:**

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

First

$$E(x^2) = \frac{1}{n} \sum_{x=1}^n x^2 = \frac{1}{n} \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}$$

Already we know

$$\mu = E(x) = \frac{n+1}{2}$$

Now Variance $\sigma^2 = E(x^2) - [E(x)]^2$

$$\sigma^2 = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 \Rightarrow \frac{n^2+2n}{4}$$

$$= \frac{4(n^2+n+2n+1) - 6(n^2+2n+1)}{24} = \frac{4(2n^2+3n+1) - 6(n^2+2n+1)}{24}$$

$$= \frac{4(n^2+n) - 6(n^2+2n+1)}{24} = \frac{4(n^2+n) - 6(n^2+2n+1)}{24}$$

$$= \frac{4(2n^2+3n+1) - 6(n^2+2n+1)}{24} = \frac{8n^2+12n+4 - 6n^2-12n-6}{24}$$

$$= \frac{2n^2-2}{24} \Rightarrow \frac{2(n^2-1)}{24} = \frac{n^2-1}{12}$$

$$\sigma^2 = \frac{n^2-1}{12} \rightarrow \text{M.C.Q}$$

$$\sigma = \sqrt{\frac{n^2-1}{12}}$$

Standard Deviation: (d.u.d.) $\rightarrow$ discrete uniform distribution

$$\sigma = \sqrt{\frac{n^2-1}{12}} \quad \text{M.C.Q}$$

Ques

Example: let us consider the rolling of a fair die, when X is representing the number of dots on the uppermost face of the die, the values would be 1, 2, 3, 4, 5, 6. $\therefore$ therefore $n=6$

Sol:

let $n=6$

$$\text{Mean} = \mu = E(x) = \frac{n+1}{2} = \frac{6+1}{2} = \frac{7}{2}$$

$$\text{mean} = \mu = 3.5$$

Topic # 119

Derivation of the MGF of the Discrete Uniform Distribution.

$\Rightarrow$ A discrete uniform distribution defined on the set $\{a, a+1, a+2, \dots, b-1, b\}$ is defined as denoted by the symbol $U(a, b)$.

where "a" and "b" are parameters of the discrete uniform distribution.

$\Rightarrow$ we consider the discrete uniform distribution defined on the set $\{1, 2, 3, \dots, n\}$

we have

$$P(X) = \frac{1}{n} \quad \text{M.C.Q.}$$

Formula MGF.

$$M(t) = E(e^{tx}) = \sum_{x=1}^n \left(e^{tx} \cdot \frac{1}{n} \right) \quad \text{M.C.Q.}$$

$$= \frac{1}{n} \sum_{x=1}^n (e^{tx}) = (e^t)^x$$

⇒ Sum of geometric Series with n -terms
 Sum of the Series = $\frac{a(1-r^n)}{(1-r)}$

Here $a = e^t$, $r = e^t \dots e^t \neq 1$

$$\sum_{x=1}^n (e^t)^x = \frac{e^t(1-e^{tn})}{(1-e^t)}$$

Now

$$M(t) = \frac{1}{n} \sum_{x=1}^n \frac{e^t(1-e^{tn})}{(1-e^t)} = \frac{e^t(1-e^{tn})}{n(1-e^t)}$$

Hence

$$M(t) = \frac{e^t - e^{(n+1)t}}{n(1-e^t)} \quad \text{--- M.C.Q}$$

Topic # 120

(Binomial distribution)

⇒ Binomial experiment:

An experiment is called a binomial experiment if it possesses the following four properties.

1) - The outcomes of each trial may be two categories (i) Success (S) (ii) failure (F)
 (outcome of interest a success or the other)

2) - The probability of Success, denoted by (P), remains constant for all trials
 (P = 1/2, 1/3, 1/4, ...)

3) - The successive trials are all independent

fix method

(الطريقة الثابتة) - في التجربة المتكررة

4) The experiment is repeated a fixed number of times, say n .

Formula: Binomial Distribution

PMF of X is given by

$$P(X) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x=0,1,\dots,n \\ 0 & \text{elsewhere} \end{cases}$$

$\Rightarrow$ where p is probability of success in each trial

$\Rightarrow$ " n " is the number of trials.

0 elsewhere

$\Rightarrow$ where p is probability of Success in each trail,

$\Rightarrow$ "n" is the number of trails.

Topic # 121

Imp Example: An event has the Probability $p = 3/8$. find the complete binomial distribution for $n = 5$ trials.

Sol. let $p = \frac{3}{8} \Rightarrow q = 1 - p$
 $q = 1 - \frac{3}{8} \Rightarrow \frac{8-3}{8} = \frac{5}{8}$

$$n = 5$$

$$x \quad P(x) = \binom{n}{x} p^x q^{n-x}$$

$$0 = \binom{5}{0} \left(\frac{3}{8}\right)^0 \left(\frac{5}{8}\right)^{5-0}$$

$$1 = \binom{5}{1} \left(\frac{3}{8}\right)^1 \left(\frac{5}{8}\right)^{5-1}$$

$$\binom{5}{0} = \frac{5!}{0!(5-0)!} = \frac{5!}{5!} = 1$$

Put $x=0$

$$p(0) = \binom{5}{0} \left(\frac{3}{8}\right)^0 \left(\frac{5}{8}\right)^5$$

$$= \left(\frac{5}{8}\right)^5 = \frac{5 \times 5 \times 5 \times 5 \times 5}{8 \times 8 \times 8 \times 8 \times 8} = \frac{3125}{32768}$$

$$= 0.09536$$

put $x=1$

$$p(1) = \binom{5}{1} \left(\frac{3}{8}\right)^1 \left(\frac{5}{8}\right)^4$$

$$= \frac{5 \cdot 3}{8} \times \frac{5 \times 5 \times 5 \times 5}{8 \times 8 \times 8 \times 8} = \frac{5 \times 3 \times 5 \times 5 \times 5 \times 5}{8 \times 8 \times 8 \times 8 \times 8}$$

$$= \frac{9375}{32768} = 0.28610$$

Similarly for $x=2, x=3, x=4$ and $x=5$

Now

x	0	1	2	3	4	5
$p(X=x)$	0.0954	0.2861	0.3333	0.2060	0.0618	0.0074

The sum of all the probability is equals to 1.

Proof

Binomial Theorem

$$(q+p)^n = \sum_{x=0}^n \binom{n}{x} p^x q^{n-x}$$

The desired probabilities are the successive terms in the binomial expansion

Sum
0.998

Topic # 122
 The shape of the binomial probability distribution depends on the value of two parameters (p) and (q)

when $p < \frac{1}{2}$ (distribution is positively skewed)
 when $p > \frac{1}{2}$ (distribution is negatively skewed)
 when $p = \frac{1}{2}$ (distribution is always symmetric)

For large n ,
 binomial probability distribution is symmetrical and mesokurtic

Topic # 123

Example If Y is $b(n, \frac{1}{3})$, then $P(Y \geq 1) = 1 - P(Y = 0) = 1 - (\frac{2}{3})^n$
 Find the smallest value of n that yields $P(Y \geq 1) \geq 0.80$.

Sol. Let given by

$$P(Y \geq 1) = 1 - P(Y \leq 1) = 1 - P(Y = 0) = 1 - (\frac{2}{3})^n$$

we know that

$$P(Y = y) = \binom{n}{y} p^y (1-p)^{n-y} \text{ or } q^{n-y} \quad \alpha = 1-p$$

Put $y = 0$

$$P(Y = 0) = \binom{n}{0} p^0 (1-p)^n$$

Here $p = \frac{1}{3}$ and n

so,

$$P(Y = 0) = \left(1 - \frac{1}{3}\right)^n = \left(\frac{2}{3}\right)^n$$

we required

$$P(Y \geq 1) \geq 0.80$$

$$(1 - \frac{2}{3})^n > 0.80$$

$$1 - 0.80 > (\frac{2}{3})^n$$

$$0.20 > (\frac{2}{3})^n$$

The condition as $(1 - \frac{2}{3})^n > 0.80$ implying
that $0.20 > (\frac{2}{3})^n$
let

$$0.20 > (\frac{2}{3})^n$$

Taking log of both sides
natural log

$$\ln(0.20) > n \ln(\frac{2}{3})$$

$$-1.6094 > n(-0.4055)$$

$$+1.6094 > +0.4055n$$

$$1.6094 < 0.4055n$$

$$n > \frac{1.6094}{0.4055} \quad \text{ie } n > 3.97$$

- is exp.
change
0.80
greater than
and then

we see that $n=4$ is the solution

M.C.Q (The probability is greater than 0.80 ie 80%
at least one success is obtained $n=4$
the probability $P = \frac{1}{3}$)

Topic # (124)

Mean and Variance of the Binomial
distribution explained through example.

Example: The Probability that a planted
radish seed germinates is 0.80. A
gardener plants nine seeds.

Let x denote the number of ^{radish} seeds that
successfully germinate.

1) what is the average number

Most Important Notes / Questions for final

of seeds the gardener could expect to germinate?

2) what is the standard deviation of X

3) what is the probability that, out of the nine seeds, at least 8 will germinate?

Sol (1)

Ans: Let, The gardener could expect, on average $\frac{9 \times 0.80}{n \times p} = 7.2$ seeds to germinate

(2)

Ans: If X is a binomial random variable, then we know that the variance is

$$np(1-p) = 9 \times 0.80 \times 0.2 = 1.44$$

$\left| \begin{array}{l} 1-p \\ 1-0.80 \\ = 0.20 \end{array} \right.$

The standard deviation of X is $= 1.44$
or 1.20

(3)

Ans: we know that

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \binom{9}{x} (0.80)^x (0.20)^{9-x}$$

so $P(X=8)$, $P(X=9)$

$$P(X=8) = \binom{9}{8} (0.80)^8 (0.20)^1$$

$$= \frac{9!}{8!(9-8)!} (0.16777216) (0.20)$$
$$= \frac{9 \times 8!}{8! \times 1}$$

$$= 9(0.16777216)(0.20) = 0.3020$$

$$p(x=9)$$

$$= \binom{9}{9} (0.80)^9 (0.20)^0$$

$$= 1 (0.1342) (1)$$

$$p(x=9) = 0.1342$$

$$= 0.3020 + 0.1342$$

$$= 0.4362$$

$$= 0.4362 \times 100\%$$

$$= 43.62\% \quad (\text{a non-trivial amount of probability})$$

Topic # 125

$\Rightarrow$ Derivation of the MGF of a Binomial distribution:

The mgf of a binomial distribution is

$$M(t) = \sum_x e^{tx} p(x) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \sum_{x=0}^n \binom{n}{x} (e^t p)^x (1-p)^{n-x}$$

$$\Rightarrow M(t) = [(1-p) + pe^t]^n$$

for all real values of t .

Now, the Mean (μ) and Variance (σ^2) is compute from $M(t)$.

$$\Rightarrow \text{we know that } \mu = M'(0)$$

$$\Rightarrow \sigma^2 = M''(0) - \mu^2$$

$$\text{let } M(t) = [(1-p) + pe^t]^n$$

Taking derivative both sides

$$M'(t) = n[(1-p) + pe^t]^{n-1} (pe^t) \rightarrow A$$

Taking again derivative of both side

$$M''(t) = n[(1-p) + pe^t]^{n-1} (pe^t) + n(n-1)[(1-p) + pe^t]^{n-2} (pe^t)^2 \rightarrow B$$

put $t=0$ in eq. (A)

$$M'(0) = n[(1-p) + pe^0]^{n-1} (pe^0)$$

$$= n[(1-p) + p \cdot 1]^{n-1} (p \cdot 1)$$

$$\mu = M'(0) = n[(1-p) + p]^{n-1} (p)$$

$$= n[1-p+p]^{n-1}$$

$$\mu = n[1]^{n-1}(p) = np$$

put $t=0$ in eq. (B)

$$M''(t) = n[(1-p) + pe^0]^{n-1} (pe^0) + n(n-1)[(1-p) + pe^0]^{n-2} (pe^0)^2$$

$$= n[(1-p) + p]^{n-1} (p) + n(n-1)[(1-p) + p]^{n-2} (p)^2$$

$$= n[1-p+p]^{n-1} (p) + n(n-1)[1-p+p]^{n-2} p^2$$

$$= n[1]^{n-1} (p) + n(n-1)[1]^{n-2} p^2$$

$$\sigma^2 = M''(0) - \mu^2$$

Now

$$\sigma^2 = M''(0) - \mu^2$$

$$\sigma^2 = np + n(n-1)p^2 - (np)^2$$

$$= [np + n(n-1)p^2] - (np)^2$$

$$= np + (n^2 - n)p^2 - (np)^2$$

$$\begin{aligned} (i) \mu &= np \\ (ii) \sigma^2 &= np(1-p) \end{aligned} \rightarrow \underline{\underline{M.C.Q}}$$

$$\begin{aligned} \sigma^2 &= np + np - np^2 - np^2 \\ &= np - np^2 \Rightarrow np(1-p) \\ \sigma^2 &= np(q) \end{aligned} \quad \left| \begin{array}{l} \therefore q = 1-p \end{array} \right.$$

Topic # 126

Example: If the mgf of a Random Variable X is

$$M(t) = \left(\frac{2}{3} + \frac{1}{3} e^t \right)^5$$

Thus, X has a binomial distribution with $n=5$ and $p=1/3$

Sol. Let, the Pmf is

$$P(x) = \begin{cases} \binom{5}{x} \left(\frac{1}{3} \right)^x \left(\frac{2}{3} \right)^{5-x} & x=0,1,2,\dots \\ 0 & \text{elsewhere} \end{cases}$$

$\Rightarrow$ Now we find Mean.

$$\begin{aligned} \mu &= np \\ &= 5 \times \frac{1}{3} = \frac{5}{3} \end{aligned}$$

$\Rightarrow$ Now, we find Variance.

$$\sigma^2 = np(1-p)$$

$$\begin{aligned} 1-p &= q \\ q &= 1 - \frac{1}{3} = \frac{3-1}{3} = \frac{2}{3} \end{aligned}$$

$$= 5 \times \frac{1}{3} \left(\frac{2}{3} \right) = 5 \times \frac{2}{9} = \frac{10}{9}$$

Topic # 127

Theorem: Let $X_1, X_2, \dots, X_m$ be independent Random variables such that X has binomial

$b(n, p)$ distribution. for $i = 1, 2, \dots, m$

$$Y = \sum_{i=1}^m X_i$$

Then Y has a binomial $b\left(\sum_{i=1}^m n_i, p\right)$ distribution

Proof:

The mgf of a binomial random variable of X

$$M_X(t) = (1 - p + pe^t)^n$$

Since $X_i, i = 1, 2, 3, \dots, m$

with $p_1 = p_2 = p_3 = \dots = p_m = p$ (constant)

So by independence

$$M_{X_i}(t) = (1 - p + pe^t)^{n_i}$$

Now

$$M_Y(t) = \prod_{i=1}^m (1 - p + pe^t)^{n_i} = (1 - p + pe^t)^{\sum_{i=1}^m n_i}$$

(Sum)

$$= (q + pe^t)^{\sum_{i=1}^m n_i}$$

Hence

$Y = \sum_{i=1}^m X_i$ has a binomial

$$b\left(\sum_{i=1}^m n_i, p\right)$$

Topic # 128

Negative Binomial Distribution

$\Rightarrow$ Consider Bernoulli trials

1) There are two possible outcomes

2) The trials are independent

3) p , the probability of success, remains the same from trial to trial.

Short Ques

For r^{th} Success, where r is fixed integer.
Then the Probability mass function of X is

$$P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, r+2, \dots$$

and

we say that X has a negative binomial (r, p) distribution

Example: Suppose that a game consists of tossing a fair coin until we obtain the 3rd head.

Then, head is "Success".

$$p = P(H) = 0.5 \quad \text{and} \quad r = 3$$

جس کو 3rd head
intended
Success

Sol:

By using negative binomial distribution

$$P(X=x) = \binom{x-1}{3-1} (0.5)^3 (1-0.5)^{x-3}, \quad x = 3, 4, 5, \dots$$

or

$$P(X=x) = \binom{x-1}{2} (0.5)^3, \quad x = 3, 4, 5, \dots$$

Put $x = 3$

$$P(3) = \binom{3-1}{2} (0.5)^3 = \binom{2}{2} (0.5)^3 = (1)(0.125) = 0.1250$$

Put $x = 4$

$$P(4) = \binom{4-1}{2} (0.5)^4 = \binom{3}{2} (0.5)^4$$

Similarly remaining part

$$P(4) = \frac{3!}{2!(3-2)!} = \frac{3 \times 2!}{2!} = 3(0.625) = 0.1875$$

$\Rightarrow$ Mean and Variance of the negative Binomial distribution.

(i) **Mean:** The Mean of negative binomial distribution is

$$E(X) = \frac{r(1-p)}{p} \quad \text{---} \quad \text{M.C.Q.}$$

(ii) **Variance:** The Variance of negative binomial distribution is

$$\text{Var}(X) = \frac{r(1-p)}{p^2} \quad \text{---} \quad \text{M.C.Q.}$$

(iii) **Standard deviation:** The Standard deviation of negative binomial distribution is

$$\text{Standard dev.} = \sqrt{\frac{r(1-p)}{p^2}} \quad \text{or} \quad \frac{\sqrt{r(1-p)}}{p}$$

(iv) **MGF of negative binomial distribution:**

$$M(t) = \left(\frac{1-p}{1-pe^t} \right)^r \quad \text{for } t < -\log p \quad \text{---} \quad \text{M.C.Q.}$$

(v) **Geometric distribution:**

$$P(X=x) = \binom{x-1}{0} p^1 (1-p)^{x-1}, \quad x=1, 2, 3, \dots \quad \text{---} \quad \text{M.C.Q.}$$

5 marks 2022

Topic # 129

Example: An oil company conducts a geological study that indicates that an exploratory oil well should have a 20% chance of striking oil. The company is wanting ^{Success} twice i.e. two oil wells from where they will be able to pull out oil and is willing to try up to five times?
What is the probability that the second success comes on the fifth we drilled?

Sol. As we know that

$$P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r} \quad x=r, r+1, r+2, \dots$$

$$\text{Here } p = 0.20 \quad r = 2$$

$$\text{Put } r = 2 \quad p = 0.20$$

$$\begin{aligned} q &= 1-p \\ &= 1-0.20 \\ q &= 0.80 \end{aligned}$$

$$P(X=x) = \binom{x-1}{2-1} (0.20)^2 (0.80)^{x-2}, \quad x=2, 3, 4, \dots$$

$$P(X=x) = \binom{x-1}{1} (0.20)^2 (0.80)^{x-2}, \quad x=2, 3, 4, \dots$$

$$P(X=x) = (x-1) (0.20)^2 (0.80)^{x-2}, \quad x=2, 3, 4, \dots$$

$$\text{Put } x = 5$$

$$P(X=5) = (5-1) (0.20)^2 (0.80)^{5-2}$$

$$= (4) (0.20)^2 (0.80)^3$$

$$= 0.082 \Rightarrow 0.082 \times 100\%$$

$$P(X=5) = 8.2\%$$

$\Rightarrow$ Let X denote the number of trials until the first Success, (head).
 The Probability mass function of X is
 $\Rightarrow P_X(x) = (1-p)^{x-1} \cdot p, \quad x = 1, 2, 3, \dots$

Topic # 130

Example: Suppose that a game consists of tossing a fair coin until we obtain the first head.

Then $p = P(H) = 0.5$

What is the geometric distribution in this case?

Solⁿ

As we know that

$$P_X(x) = (1-p)^{x-1} p, \quad x = 1, 2, 3, \dots$$

$p = 0.5$

$$P_X(x) = (1-0.5)^{x-1} (0.5) = (0.5)^{x-1} (0.5) = (0.5)^x \quad x = 1, 2, 3, \dots$$

$$P_X(x) = \frac{1}{2^x} \quad x = 1, 2, 3, \dots$$

put $x = 1$

$$P_X(1) = \frac{1}{2^1} = \frac{1}{2} = 0.5$$

put $x = 2$

$$P_X(2) = \frac{1}{2^2} = \frac{1}{4} = 0.25$$

put $x = 3$

$$P_X(3) = \frac{1}{2^3} = \frac{1}{8} = 0.125$$

Similarly remain $x = 4, 5, 6, \dots$

Topic # 131

Example: If the Probability that a person will believe a rumor about the retirement of a certain politician is 0.25, what is the probability the sixth person to hear the rumor will be the first to believe it.

جو کر رہا ہے لڑوہ Success ہوا اور جو نہیں لڑا ہے وہ Failure ہو گیا
 Question یا Example میں کسی سے کہہ کر اس میں Bernoulli Trial (Bernoulli Trial) استعمال کریں
 (Success, Failure)

Sol. The Probability mass function (PMF) of x is
 $P_x(x) = (1-p)^{x-1} p, \quad x=1, 2, 3, \dots \rightarrow (1)$

$\Rightarrow$ Here $p = 0.25$

Ans

$$P_x(x) = (1-0.25)^{x-1} (0.25) = (0.75)^{x-1} (0.25) \quad x=1, 2, 3, \dots$$

$\Rightarrow$ By given condition put $x=6$

$$\begin{aligned} P_x(6) &= (0.75)^{6-1} (0.25) = (0.75)^5 (0.25) \\ &= 0.2373 \times 0.25 \\ &= 0.059 \Rightarrow 0.059 \times 100\% \end{aligned}$$

$$P(x=6) = 5.9\%$$

$\Rightarrow$ If the first person to believe the rumor

put $x=1$

$$\begin{aligned} P(x=1) &= (0.75)^{1-1} (0.25) \\ &= 1 \times 0.25 = 0.25 \\ &= 0.25 \times 100\% = 25\% \end{aligned}$$

Topic # 132

Mean and Variance of Geometric Distribution

Example: A representative from an ice-cream producing company randomly stops people on a randomly selected street in the city of Lahore until he finds a person who actually purchased and tried the latest product of this company meaning a new kind of ice-cream.

Q = What is the Variance

Sol: for geometric random variable X is,
 $\therefore \mu = E(X) = \frac{1}{p} = \frac{1}{0.20} = 5$

For variance of geometric random variable X is

$$\sigma^2 = \text{Var}(X) = \frac{1-p}{p^2}$$

$$\sigma^2 = \frac{1-0.20}{(0.20)^2} = \frac{0.80}{0.04} = 20$$

$$\sqrt{\sigma^2} = \sqrt{20} \Rightarrow \sqrt{4 \times 5} \Rightarrow \sqrt{4} \times \sqrt{5}$$

Standard deviation $\Leftarrow \sigma = 2\sqrt{5}$

Topic # 133

Concept of Multinomial Distribution

$\Rightarrow$ The binomial distribution is generalized to be multinomial distribution

$\Rightarrow$ Let a random experiment be repeated 'n' independent times

$\Rightarrow$ on each repetition, the experiment results is one of K mutually exclusive and exhaustive outcomes, say, $C_1, C_2, C_3, \dots, C_K$

$\Rightarrow$ Let p_i represent the probability of outcome is C_i and $i = 1, 2, \dots, K$

$\Rightarrow$ For random variable X_i of outcome C_i
 $i = 1, 2, 3, \dots, K-1$

Let $x_1, x_2, \dots, x_{K-1}$ be non-negative integers
 $x_1 + x_2 + \dots + x_{K-1} \leq n$

2.9 Suppose that a die is tossed 30 times.

$\Rightarrow X_1$ represents the event how many times 1 comes

$\Rightarrow$ " " " " " " " "

Let $K = 6$

$$X_1 = 2, X_2 = 3, X_3 = 4, X_4 = 5, X_5 = 6 \rightarrow \text{M.C.Q.} \text{ } \textcircled{3}$$

By adding

$$X_1 + X_2 + X_3 + X_4 + X_5 = n$$

$$2 + 3 + 4 + 5 + 6 = 20$$

$$\begin{array}{r} \text{Total} = 30 \\ \text{Adding} = 20 \\ \hline \text{rem} = 10 \\ \text{Ans} \end{array}$$

So, the last probability 10 times to

Come $X_6 = 10$

Hence

$$\begin{aligned} &= n - (X_1 + \dots + X_{K-1}) \quad \text{M.C.Q.} \\ &= 30 - (2 + 3 + 4 + 5 + 6) \\ &= 30 - 20 = 10 \end{aligned}$$

The probability mass function (pmf) of multinomial distribution is

$$f(X_1, \dots, X_K, n, P_1, \dots, P_K)$$

M.C.Q.

$$\begin{cases} \frac{n!}{x_1! x_2! \dots x_K!} P_1^{x_1} P_2^{x_2} \dots P_K^{x_K} & \text{when } \sum_{i=1}^K x_i = n \\ 0 & \text{otherwise} \end{cases}$$

for non-negative integers $x_1, \dots, x_K$

For Binomial distribution

$$\text{Mean} = E(X) = np$$

$$\text{Variance} = \text{Var}(X) = np(1-p)$$

=> For Mean: (Multinomial distribution)

The Mean of the Multinomial distribution is

$$E(X_i) = np_i \quad \text{M.C.Q.}$$

=> For Variance:

The Variance of the multinomial distribution is

$$\text{Var}(X_i) = np_i(1-p_i) \quad \text{M.C.Q.}$$

=> For Covariance:

The covariance of the multinomial distribution is

$$\text{Cov}(X_i, X_j) = -np_i p_j \quad \text{M.C.Q.} \quad i \neq j$$

Topic # 134

The Multinomial distribution

let

$$f(x_1, x_2, \dots, x_k, n, p_1, p_2, \dots, p_k) = \begin{cases} \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k} & \text{when } \sum_{i=1}^k x_i = n \\ 0 & \text{otherwise} \end{cases}$$

put $k=2$

$$\sum_{i=1}^2 x_i = n \quad \text{means} \quad x_1 + x_2 = n \Rightarrow x_2 = n - x_1$$

$$\sum_{i=1}^2 p_i = 1 \quad \text{means} \quad p_1 + p_2 = 1 \Rightarrow p_2 = 1 - p_1$$

$$f(x_1, x_2, n, p_1, p_2) = \frac{n!}{x_1! x_2!} p_1^{x_1} p_2^{x_2} \quad \text{when } \sum_{i=1}^2 x_i = n$$

(A)

we re-write the equation (A)

$$f(x_1, n, p_1) = \frac{n!}{x_1!(n-x_1)!} p_1^{x_1} (1-p_1)^{n-x_1}$$

V.V. Imp

$$f(x; n, p) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

Binomial distribution
 $x = 0, 1, 2, \dots, n$

Topic # 135 Multinomial

Example: Suppose that the racial/ethnic distribution in a very large western city is given by the table that follows

white	Black	Hispanic
65%	20%	15%

- July
 what is the probability
- Seven white, three black, and two Hispanic
 - Four white, and eight other members
 - At the most, one black member

Sol: (i) - what is the probability

that the July contains seven white, three Black, and two Hispanic members?

Sol: Let $K = 3$ (Trinomial distribution) $\Rightarrow$ 3 possibilities

Consider the vector $X = (x_1, x_2, x_3)$

x_1 = number of white members = (7)

x_2 = number of Black members = (3)

x_3 = number of Hispanic members = (2)

Here

$$x_1 = 7, x_2 = 3, x_3 = 2$$

$$\begin{aligned} n &= 7+3+2 \\ &= 12 \\ \text{white} &= 65\% = 0.65 \\ &= \frac{65}{100} \\ &= 0.65 \end{aligned}$$

$$\begin{aligned} P(x_1=7, x_2=3, x_3=2) &= \frac{n!}{x_1! x_2! x_3!} p_1^{x_1} p_2^{x_2} p_3^{x_3} \\ &= \frac{12!}{7! 3! 2!} (0.65)^7 (0.20)^3 (0.15)^2 \end{aligned}$$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{7! 3!} (0.0490222) (0.008) (0.0025)$$

$$= 0.0699 \times 100$$

$$= 6.99\% \text{ or } 7\%$$

(ii) - what is the probability that the jury contains four white and eight other members? (6, 12, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1)

Sol.

Let $K = 2$

$$X_1 = 4$$

and

$$X_2 = 8$$

$$f(X_1 = 4, X_2 = 8) = \frac{12!}{4! 8!} (0.05)^4 (0.10)^8$$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8!}{4 \times 3 \times 2 \times 1 \times 8!} ()$$

(iii)

Let $K = 2$

Ans

$$X_1 = 0$$

and

$$X_2 = 12$$

$$f(x, n, p) = \frac{n!}{x! (n-x)!} p^x (1-p)^{n-x}$$

$$P(X=0) = \frac{12!}{0! (12-0)!} (0.20)^0 (0.80)^{12}$$

$$= (1) (0.687194)$$

$$= 0.687194$$

and

$$P(X=1) = \frac{12!}{1! (12-1)!} (0.20)^1 (0.80)^{11}$$

$$= \frac{12 \times 11}{1!} (0.20) (0.0858993)$$

$$p = 0.20$$

$$q = 1 - p$$

$$= 1 - 0.20$$

$$q = 0.80$$

$$= 0.2061584 \dots$$

let

$$\begin{aligned} P(X=0 \text{ or } X=1) &= P(X=0) + P(X=1) \\ &= 0.069 + 0.206 \\ &= 0.275 \times 100 \\ &= 27.5\% \end{aligned}$$

white	Black and Hispanic
65%	20% + 15% = 35%

(ii)

$$\text{let } n=2$$

$$\text{Ans: } x_1=4, x_2=8$$

$$\begin{aligned} p &= 65\% \\ p &= 1 - 0.35 \\ n &= 0.35\% \end{aligned}$$

$$\begin{aligned} f(x_1=4, x_2=8) &= \frac{12!}{4!(12-4)!} (0.65)^4 (0.35)^8 \\ &= \frac{12!}{4!8!} (0.17850625) (0.0002251875) \\ &= 495 (0.17851) (0.00023) \\ &= 0.0199 \times 100 \\ &= 1.99\% \text{ or } 2\% \end{aligned}$$

Topic # 136

Moment Generating Function (MGF) of Multinomial Dist.

Let $X = (X_1, X_2, \dots, X_k)$ be a random vector.

$$\text{If } E(e^{t_1 X_1 + t_2 X_2 + \dots + t_k X_k}) \text{ exists for } |t_1| < h_1, |t_2| < h_2, \dots, |t_k| < h_k$$

where $h_1, h_2, \dots, h_k$ are positive then $E(e^{t_1 X_1 + t_2 X_2 + \dots + t_k X_k})$ is denoted by $M_{X_1, X_2, \dots, X_k}(t_1, t_2, \dots, t_k)$ the moment generating function.

$\Rightarrow$ The MGF of Multinomial distribution is

$$M_X(t) = M_X(t_1, t_2, \dots, t_k) = E\left(e^{\sum_{i=1}^k t_i X_i}\right)$$

$$M_X(t) = \left(\sum_{i=1}^k p_i e^{t_i}\right)^n \quad \text{M.C.O.}$$

$\Rightarrow$ The cumulant generating function is

$$K_X(t) = K_X(t_1, t_2, \dots, t_k)$$

$$= \log [M_X(t_1, t_2, \dots, t_k)]$$

$$K_X(t) = n \log \left(\sum_{i=1}^k p_i e^{t_i}\right) \quad \text{M.C.O.}$$

eg $k=3$

$n=2$

$$= \left(\sum_{i=1}^3 p_i e^{t_i}\right)^2 = (p_1 e^{t_1} + p_2 e^{t_2} + p_3 e^{t_3})^2$$

$$(p_1 e^{t_1} + p_2 e^{t_2} + p_3 e^{t_3})^2 = p_1^2 e^{2t_1} + p_2^2 e^{2t_2} + p_3^2 e^{2t_3} + 2p_1 p_2 e^{t_1+t_2} + 2p_1 p_3 e^{t_1+t_3} + 2p_2 p_3 e^{t_2+t_3}$$

$$E(X_1) = \frac{\partial}{\partial t_1} (M_{X_1, X_2, X_3}(0, 0, 0))$$

$$E(X_2) = \frac{\partial M}{\partial t_2} (0, 0, 0)$$

$$E(X_3) = \frac{\partial M}{\partial t_3} (0, 0, 0)$$

Topic # 137 M.C.O.

Hypergeometric distribution

(i) eg (pass/fail or Employed/unemployed)

(ii) The successive draws are not independent of each other.

(iii) The number of draws is fixed in advance.

Example: Suppose That we have a class of PhD students in a small educational institution in some particular subject and the total number of PhD students of that subject is 5.
So, $N=5$

We would like to draw 3 students to send on a conference then $n=3$.

Q1 Taking the sample of size 3 without replacement
Let Total student $N=5$

A Grade student = 2

$$\text{probability} = \frac{2}{5}$$

If we select one student $= N-1 = 4$
So

The probability of remain student = $\frac{1}{4}$
for grade A

Hypergeometric distribution:

$$P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad x=0,1,2,\dots,K \text{ if } K \leq n$$

M.C.Q

$$\text{and } x=0,1,2,\dots,n \text{ if } n < K$$

where

- (i) N = is the population size.
- (ii) K = is the number of success in the population at the beginning of the experiment.
- (iii) n = is the number of draws (quantity drawn in each trial)
- (iv) x = is the number of observed success

Topic # 138

Example: The name of 5 men and 5 women are written on slips of paper and placed in a hat. Four names are drawn.

What is the probability that 2 are men and 2 are women?

Sol: Let X denote the number of men.
Then

$$N = 5 + 5 = 10 \text{ names to be drawn from}$$

$$K = 5, \quad n = 4, \quad x = 0, 1, 2, 3, 4$$

As we know that

$$h(x; 10, 4, 5) = \frac{\binom{5}{x} \binom{10-5}{4-x}}{\binom{10}{4}} = \frac{\binom{5}{x} \binom{5}{4-x}}{\binom{10}{4}}$$

Let $P(X=2)$ by required

$$h(2; 10, 4, 5) = \frac{\binom{5}{2} \binom{5}{2}}{\binom{10}{4}} = \frac{5!}{2!(5-2)!} \cdot \frac{5!}{2!(5-2)!} \cdot \frac{1}{10!}$$

$$= \frac{5 \times 4 \times 3!}{2! 3!} \cdot \frac{5 \times 4 \times 3!}{2! 3!} = \frac{20 \times 20}{2 \times 2} = \frac{5040}{24}$$

$$= \frac{10 \times 10}{2! 0} = \frac{100}{2!} = \frac{10}{2}$$

$\Rightarrow$ For Mean (of Hypergeometric distrib.)

$$\mu = np$$

$\Rightarrow$ For Variance

$$\sigma^2 = npq \left(\frac{N-n}{N-1} \right)$$

$$\left(p = \frac{K}{N}, q = \frac{N-K}{N} \right)$$

Derivation of Mean
#139

$$E(X) = \frac{K}{N} \sum_{x=0}^N \binom{N-K}{x-1} \binom{N-K}{n-x}$$

$$= \frac{K}{N} \binom{N-1}{n-1} \left[\sum_{j=0}^n \binom{a}{j} \binom{b}{n-j} \right]$$

$$\left[\sum_{j=0}^n \binom{a}{j} \binom{b}{n-j} = \binom{a+b}{n} \right]$$

Let $y = x-1$
 $x = y+1$
 where $x=1, y=0$
 $x=n, y=n-1$

Topic # 140

Derivation of the Variance of the Hypergeometric distribution

M.C.Q

$$E(X^2) = \frac{nK}{N} + \frac{K(K-1)}{\binom{N}{n}} \binom{N-2}{n-2} \left[\sum_{j=0}^n \binom{a}{j} \binom{b}{n-j} \binom{a+b}{n} \right]$$

$$\sigma^2 = \text{Var}(X) = E(X^2) - \mu^2$$

$$= \frac{nK}{N} + \frac{K(K-1)}{\binom{N}{n}} \binom{N-2}{n-2} - \left(\frac{nK}{N} \right)^2$$

$$= \frac{nK}{N} + \frac{K(K-1)n(n-1)}{N(N-1)} - \left(\frac{nK}{N} \right)^2$$

$$\sigma^2 = \frac{nK(N^2 - nN - KN + nK)}{N^2(N-1)}$$

M.C.Q

$$\sigma^2 = \frac{nK(N-K)}{N^2} \cdot \frac{N-n}{(N-1)}$$

$$\begin{aligned} N^2 - nN - KN + nK &= N^2 - nN - nK + nK \\ &= N^2 - nN - nK + nK \\ &= N(N-K) - n(N-K) \\ &= (N-K)(N-n) \end{aligned}$$

So

$$\text{Var}(X) = npq \cdot \frac{N-n}{N-1}$$

M.C.Q

$$p = \frac{K}{N}, \quad q = \frac{N-K}{N}$$

M.C.Q

- Q. 1. it appears as the limiting form of the binomial distribution under the certain conditions.
 2. when we apply Poisson distribution in that real-life situation where we are dealing with Poisson p.c.

Topic # 141

Poisson distribution (PMF and shape distn)

⇒ consider the function $p(x)$ is defined

by

$$p(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & x = 0, 1, 2, \dots, \lambda > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where $m > 0$.

Since $\lambda > 0$

Therefore $p(x) \geq 0$

and

$$\sum_x p(x) = \sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

⇒ Recall the series,

$$1 + m + \frac{m^2}{2!} + \frac{m^3}{3!} + \dots = \sum_{x=0}^{\infty} \frac{m^x}{x!}$$

converges for all values of m to e^m .

⇒ A random variable that has a PMF of the form $p(x) = \sum_{x=0}^{\infty} \frac{m^x}{x!}$ is called a Poisson distribution.

Topic # 142

Poisson Process: A Process in which events occur randomly in a specified duration of time or in a specified portion of one-dimensional, two-dimensional or three-dimensional space (λ occurrences in one unit of time)

⇒ The probability of having x occurrences

$$P(X=x) = \begin{cases} \frac{e^{-\lambda t} (\lambda t)^x}{x!} & x = 0, 1, 2, \dots, \lambda > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Topic # 143

Example: If it is known that, on the average two minor accidents occur at a busy crossing per day, what is the probability that, at the most two minor accidents will occur during an entire week.

sol. Let $\lambda = 2$ per day
 one week = 7 days $\Rightarrow t = 7$
 $\lambda t = 2 \times 7 = 14$

So,

$$P(X=x) = \begin{cases} \frac{e^{-\lambda t} (\lambda t)^x}{x!} = \frac{e^{-14} (14)^x}{x!} & x=0, 1, 2, \dots \rightarrow (1) \\ 0 & \text{elsewhere} \end{cases}$$

Now $X < 2$:

$$P(X < 2) = P(X=0, X=1 \text{ or } X=2)$$

$$\begin{aligned} &= \frac{e^{-14} (14)^0}{0!} + \frac{e^{-14} (14)^1}{1!} + \frac{e^{-14} (14)^2}{2!} \\ &= e^{-14} + 14e^{-14} + 98e^{-14} \\ &= 113e^{-14} \\ &= 113 \times \frac{1}{e^{14}} \Rightarrow 113 \times \frac{1}{1202592.891} = 0.00009396 \end{aligned}$$

$$P(X \leq 2) = 0.00009396$$

Topic # 144

$\Rightarrow$ Mean, Variance and Coefficient of Variation of the Poisson distribution.

Poisson distribution, mean = Variance $= \lambda > 0$

Note: ① If λ increases, the coefficient of variation of Poisson distribution decreases.

② If λ tends to infinity, the coefficient of variation of Poisson distribution tends to zero.

A Poisson pmf is

$$P(x) = \begin{cases} \frac{\mu^x e^{-\mu}}{x!} & x=0, 1, 2, \dots \\ 0 & \text{elsewhere} \end{cases}$$

The parameter λ or μ is a Poisson distribution.

$\Rightarrow$ Coefficient of variation of Poisson distribution:

$$\text{COV} = \frac{\sigma}{\mu} \times 100\% \quad \left| \begin{array}{l} \sigma^2 = \lambda \\ \sigma = \sqrt{\lambda} \\ \mu = \lambda \end{array} \right.$$

$$\frac{\sqrt{\lambda}}{\sqrt{\lambda} \cdot \sqrt{\lambda}}$$

$$\text{COV} = \frac{\sqrt{\lambda}}{\lambda} \times 100\% = \frac{1}{\sqrt{\lambda}} \times 100\%$$

Example:

Suppose that X has a Poisson distribution with $\lambda = 4$

Then the pmf of X is

$$P(x) = \begin{cases} \frac{0.5^x e^{-0.5}}{x!} & x=0, 1, 2, \dots \\ 0 & \text{elsewhere} \end{cases}$$

find coefficient of variation.

So!

The mean of distribution is 4

The variance of distribution is 4

The coefficient of variation is,

$$\text{COV} = \frac{\sigma}{\mu} \times 100\% \text{ or } \frac{1}{\sqrt{\lambda}} \times 100$$

$$\text{COV} = \frac{1}{\sqrt{4}} \times 100\% = \frac{1}{2} \times 100\% = 50\%$$

Exmp

Sup - $\lambda = 9$
the pmf

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Topic # 145

Derivation of Mean of the Poisson distribution.

If the random variable X has a Poisson distribution is

$$P(X; \mu) = \frac{e^{-\mu} \mu^x}{x!}$$

Then

its mean and variance are given

by $E(X) = \mu$ and $\text{Var}(X) = \mu$

For Mean:

$$\text{Mean} = E(X) = \sum_{x=0}^{\infty} x \cdot P(X; \mu) \text{ where } P(X; \mu) = \frac{e^{-\mu} \mu^x}{x!}$$

$$= 0 \cdot \frac{\mu^0}{0!} e^{-\mu} + 1 \cdot \frac{\mu^1}{1!} e^{-\mu} + 2 \cdot \frac{\mu^2}{2!} e^{-\mu} + 3 \cdot \frac{\mu^3}{3!} e^{-\mu} + 4 \cdot \frac{\mu^4}{4!} e^{-\mu} + \dots$$

$$= 0 + 1 \cdot \mu e^{-\mu} + \mu^2 \cdot \frac{1}{e^{-\mu}} + \frac{\mu^3}{2} \cdot \frac{1}{e^{-\mu}} + \frac{\mu^4}{6} \cdot \frac{1}{e^{-\mu}} + \dots$$

$$= \mu e^{-\mu} + \mu^2 \cdot e^{-\mu} + \frac{\mu^3}{2!} e^{-\mu} + \frac{\mu^4}{3!} e^{-\mu} + \dots$$

$$= \mu e^{-\mu} \left[1 + \mu + \frac{\mu^2}{2!} + \frac{\mu^3}{3!} + \dots \right]$$

Recall $1 + \mu + \frac{\mu^2}{2!} + \frac{\mu^3}{3!} + \dots = \sum_{x=0}^{\infty} \frac{\mu^x}{x!}$

$$= \mu e^{-\mu} [e^{\mu}] = \mu$$

$\Rightarrow \boxed{\text{Mean} = \mu}$

$\downarrow$
M.C.Q. $\leftarrow$ Poisson distribution

Topic # 146

⇒ Derivation of the Mean of the Poisson distribution

The mean and variance are given below

$$E(X) = \mu \text{ and } \text{Var}(X) = \mu$$

Proof

By definition

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

AS

$$E[X^2 - X + X] = X \text{ both}$$

$$E(X^2) = E[X(X-1) + X] \Rightarrow E(X) + E[X(X-1)]$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{e^{-\mu} \mu^x}{x!} + \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\mu} \mu^x}{x!}$$

$$E(X^2) = \mu + e^{-\mu} \sum_{x=2}^{\infty} x(x-1) \frac{\mu^x}{x!}$$

⇒ multiply μ^{-2} put $x=2$

$$= \mu + \mu^2 e^{-\mu} \sum_{x=2}^{\infty} \frac{\mu^{x-2}}{(x-2)!}$$

$$E(X^2) = \mu + \mu^2 e^{-\mu} \sum_{x=2}^{\infty} \frac{\mu^{x-2}}{(x-2)!}$$

$$\text{let } y = x-2$$

$$E(X^2) = \mu + \mu^2 e^{-\mu} \sum_{y=0}^{\infty} \frac{\mu^y}{y!}$$

$$\mu = \frac{e^{-\mu} \mu^x}{x!}$$

$$= \mu + \mu^2 e^{-\mu} \frac{\mu^y}{y!} \Rightarrow \frac{\mu^y}{y!}, y=0, 1, 2, \dots$$

Hence

$$\text{Var}(X) = \mu + \mu^2$$

$$\Rightarrow \text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(X) = \mu + \mu^2 - \mu^2$$

$$\boxed{\text{Var}(X) = \mu}$$

Hence proved

Q.E.D.
Poisson distribution

M.G.F = Moment generating function

rule (Mean = m , Variance = m)

$$\mu = \sigma^2 = m > 0$$

Topic # 147

M.G.F of Poisson distribution.

The mgf of a Poisson distribution is

$$M(t) = \sum_x e^{tx} P(x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \frac{m^x e^{-m}}{x!} \Rightarrow e^{-m} \sum_{x=0}^{\infty} \frac{(me^t)^x}{x!} = e^{-m} e^{met}$$

$$\Rightarrow M(t) = e^{m(e^t-1)} \quad (\text{for all values of } t)$$

$$\Rightarrow M'(t) = e^{m(e^t-1)} (me^t)$$

$$\mu = M'(0) = m \quad \sigma^2 = M''(0) = M''(0) = M''(0)$$

$$\sigma^2 = M''(0) = m + m^2 = m + m^2 - m^2 = m$$

a Poisson distribution $\mu = \sigma^2 = m > 0$ M.C.Q

For Mean:

$$M'(t) = e^{m(e^t-1)} (me^t)$$

put $t = 0$

$$M'(0) = e^{m(e^0-1)} (me^0) \Rightarrow e^m m$$

$$= e^{m(0)} (m \cdot 1) \Rightarrow 1(m) = m$$

And For Variance:

$$\Rightarrow M''(t) = e^{m(e^t-1)} (me^t) + e^{m(e^t-1)} (me^t)^2$$

put $t = 0$

$$= e^{m(e^0-1)} (me^0) + e^{m(e^0-1)} (me^0)^2$$

$$= e^{m(1-1)} (m \cdot 1) + e^{m(1-1)} (m \cdot 1)^2$$

$$= e^{m(0)} (m) + e^{m(0)} (m)^2$$

$$= 1(m) + 1(m)^2$$

$$\Rightarrow M''(t) = m + m^2$$

Since

$$\sigma^2 = M''(0) - \mu^2$$

$$\sigma^2 = m + m^2 - m^2 = m$$

Topic # 14B

⇒ Binomial distribution of Poisson approximation.

The binomial distribution is

$$b(x, n, p) = \binom{n}{x} p^x q^{n-x} \quad x = 0, 1, \dots, n$$

$$= \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$= \frac{n(n-1)(n-2)\dots(n-x+1)}{x!} p^x q^{n-x}$$

$$= \frac{n(n-1)(n-2)\dots(n-x+1)}{x!} p^x q^{n-x}$$

Let $\mu = np$

⇒ $\mu = np \Rightarrow p = \frac{\mu}{n}, \quad q = 1-p = 1 - \frac{\mu}{n}$

let

$$b(x, n, p) = \frac{n(n-1)(n-2)\dots(n-x+1)}{x!} \left(\frac{\mu}{n}\right)^x \left(1 - \frac{\mu}{n}\right)^{n-x}$$

$$= \frac{\mu^x}{x!} \left[1 \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right) \right] \left(1 - \frac{\mu}{n}\right)^{n-x}$$

let $n \rightarrow \infty$ and $p \rightarrow 0$ such that $np = \mu$ (const)

$$\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right) \rightarrow \left(1 - \frac{x-1}{n}\right)$$

approximate unity

⇒ Now, $\left(1 - \frac{\mu}{n}\right)^n \Rightarrow \left[\left(1 - \frac{\mu}{n}\right)^{\frac{n}{\mu}} \right]^\mu$ multiply and divide by μ

By using (A)
 $= \frac{\mu^x}{x!} \left(1 - \frac{\mu}{n}\right)^n$
 $= \frac{\mu^x e^{-\mu}}{x!}$
 Hence proved

Let $\frac{n}{\mu} = k \Rightarrow \left(1 - \frac{\mu}{n}\right)^{\frac{n}{\mu}} = \left(1 - \frac{1}{k}\right)^k$

$\lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{k}\right)^k \right]^\mu$ (k times)

$= \lim_{k \rightarrow \infty} \left[\left(1 - \frac{1}{k}\right)^k \left(1 - \frac{1}{k}\right)^k \dots \left(1 - \frac{1}{k}\right)^k \right]$

⇒ $\lim_{n \rightarrow \infty} \left(1 - \frac{\mu}{n}\right)^n = e^{-\mu}$ M.C.E

The binomial distribution $h(x; n, p)$ where $n \rightarrow \infty, p \rightarrow 0$
 then the product of np (constant)
 Poisson distribution $P(X, \mu) = \frac{e^{-\mu} \mu^x}{x!}$ Binomial form $\binom{n}{x} p^x q^{n-x}$

Topic # 149

Example: Suppose 80% of the tires manufactured at a particular plant are defective. To illustrate the use of the Poisson approximation for the binomial, the probability of obtaining exactly one defective tire from a sample of 20.

Sol. Let

$$n = 20, \quad x = 1, \quad p = 0.08$$

$$q = 1 - p = 1 - 0.08 = 0.92$$

By using binomial formula

$$P(X=1) = \binom{20}{1} (0.08)^1 (0.92)^{19}$$

$$= \frac{20!}{1!(20-1)!} (0.08)^1 (0.92)^{19}$$

$$= \frac{20 \times 19!}{1! 19!} (0.08) (0.2051014)$$

$$P(X=1) = 0.3281$$

or $\Rightarrow$ By using Poisson formula

Let $\mu = np$

$$P(X, \mu) = \frac{e^{-\mu} \mu^x}{x!}$$

$$P(X=1) = \frac{e^{-(20)(0.08)} [(20)(0.08)]^1}{1!} = \frac{e^{-1.6} 1.6^1}{1}$$

$$P(X=1) = 0.32805$$

$$P(X=1) = 0.3281$$

Topic # 150

(Rectangular Distribution)

⇒ Def Probability Distribution Function.

The Probability density function of the continuous uniform distribution is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$$

⇒ M.C.Q Cumulative Distribution Function.

The cumulative distribution function of the continuous uniform distribution is

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases}$$

Derivation of interval $a \leq x \leq b$

$$\begin{aligned} F(x) &= \int_a^x f(x) dx = \int_a^x \frac{1}{b-a} dx \\ &= \frac{1}{b-a} \left[x \right]_a^x \Rightarrow (x-a) \Rightarrow \frac{x-a}{b-a} \end{aligned}$$

As

$$F(x) = \frac{x-a}{b-a} \Rightarrow \frac{x}{b-a} - \frac{a}{b-a}$$

+ ans
M.C.Q

$$F(x) = \left[\frac{1}{b-a} \right] x + \left[\frac{-a}{b-a} \right] \Rightarrow \text{it is a straight line}$$

As the straight line is $y = mx + c$

Topic # 151

Derivation of the mean and variance of the continuous uniform distribution.

* For Mean:

$$E(X) = \frac{(a+b)}{2} \quad \text{M.C.P.}$$

* For Variance:

$$\text{Var}(X) = \frac{(b-a)^2}{12} \quad \text{M.C.P.}$$

Proof of Expectation, or Mean:

$$\text{Let } f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$$

$$\begin{aligned} \text{let } E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x \left(\frac{1}{b-a} \right) dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{1}{b-a} \left[\frac{b^2}{2} - \frac{a^2}{2} \right] \\ &= \frac{1}{b-a} \left[\frac{b^2 - a^2}{2} \right] \Rightarrow \frac{(b-a)(b+a)}{2(b-a)} \end{aligned}$$

$$E(X) = \frac{b+a}{2} \text{ or } \frac{a+b}{2} \quad \text{Hence proved}$$

Proof of Variance:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\begin{aligned} \text{let } E(X^2) &= \int_a^b x^2 f(x) dx = \int_a^b x^2 \left(\frac{1}{b-a} \right) dx \\ &= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b \Rightarrow \frac{1}{b-a} \left[\frac{b^3}{3} - \frac{a^3}{3} \right] \Rightarrow \frac{b^3 - a^3}{3(b-a)} \end{aligned}$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$E(\bar{x}) = \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(a^2 + ab + b^2)}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

Now,

$$\text{Variance}(x) = E(x^2) - [E(x)]^2$$

$$\text{Var}(x) = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{a^2 + ab + b^2}{3} - \frac{a^2 + b^2 + 2ab}{4} = \frac{4(a^2 + ab + b^2) - 3(a^2 + b^2 + 2ab)}{12}$$

$$= \frac{4a^2 + 4ab + 4b^2 - 3a^2 - 3b^2 - 6ab}{12} = \frac{a^2 + b^2 - 2ab}{12}$$

$$\text{Var}(x) = \frac{(a-b)^2}{12} \Rightarrow \frac{a^2 - 2ab + b^2}{12}$$

$$\text{Var}(x) = \frac{(b-a)^2}{12}$$

Topic # 152

Example: Suppose in a Quiz, there are 30 Participants

A Question is given to all 30 participants and the time allowed to answer it is 25 seconds.

- Find the probability that a participant will respond within b seconds.
- How many of the 30 participants can be expected to respond within b seconds?

Sol.

The interval of probability distribution is 0 second to 25 second.

i.e. $0 \leq x \leq 25$

let $f(x) = \frac{1}{x} = \frac{1}{25-0} = \frac{1}{25}$

(i)- let $P(X < 6)$

So for Probability

$F(x) = \frac{x-a}{b-a}$

$a=0, b=25$ and $x=6$

$F(6) = \frac{6-a}{b-a} = \frac{6-0}{25-0} = \frac{6}{25}$

(ii)-

Expected frequency = Probability $\times$ the number

ex. frequency = $\frac{6}{25} \times 30 = \frac{36}{5}$

Expected frequency = 7

Topic # 153

Exponential distribution

The probability density function of exponential distribution is

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0 \end{cases}$$

Let $\lambda = \frac{1}{\theta}$
replace θ

(only +ve side)

So, it is obvious that is a single-Parameter distribution.

The Shape of PDF of exponential distribution $\lambda = 0.5$

- $f(x) = \frac{1}{0.5} e^{-\frac{x}{0.5}}$, $0 < x < \infty \rightarrow$ orange $\lambda = 0.5$
- $f(x) = \frac{1}{1} (e^{-\frac{x}{1}}) = e^{-x}$, $0 < x < \infty \rightarrow$ Purple $\lambda = 1$
- $f(x) = \frac{1}{1.5} e^{-\frac{x}{1.5}}$, $0 < x < \infty \rightarrow$ sky blue $\lambda = 1.5$

Note: In Poisson process events are occurring randomly over a time scale according to exponential distribution

⇒ CDF (Cumulative distribution function)
The cumulative distribution function of an exponential distribution is

$$F(x, \lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0 \end{cases}$$

Proof

$$F(x) = \int_{-\infty}^x f(x) dx$$

$$\begin{aligned} F(x) &= \int_0^x \lambda e^{-\lambda x} dx = - \int_0^x (-\lambda) e^{-\lambda x} dx \\ &= - \left[e^{-\lambda x} \right]_0^x = - \left[e^{-\lambda x} - e^0 \right] \\ &= - \left[e^{-\lambda x} - 1 \right] \\ F(x) &= 1 - e^{-\lambda x}, \quad x \geq 0 \end{aligned}$$

Topic # 154

⇒ Derivation of the Mean of the Exponential distribution. (The mean and standard deviation of exponential distribution are equal)

⇒ Derivation of the Mean:

Let $f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \lambda > 0$

Therefore

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

For integral:

$$\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx$$

let $u = x, \quad v = \lambda e^{-\lambda x}$
 $= -(-\lambda e^{-\lambda x})$

$$\int v dx = \int -(-\lambda e^{-\lambda x}) = -(e^{-\lambda x})$$

$$\frac{du}{dx} = \frac{d}{dx}(x) = 1$$

$$\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx$$

$$= x \cdot (-e^{-\lambda x}) - \int (1 \cdot (-e^{-\lambda x})) dx$$

$$= -x e^{-\lambda x} - \frac{1}{\lambda} \int (\lambda (-e^{-\lambda x})) dx$$

$$= -x e^{-\lambda x} - \frac{1}{\lambda} \int (-\lambda e^{-\lambda x}) dx$$

$$\lambda e^{-\lambda x} = -x e^{-\lambda x} - \frac{1}{\lambda} (e^{-\lambda x})$$

let

$$E(x) = \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx = \left[-x e^{-\lambda x} \right]_0^{\infty}$$

$$= \left[-x e^{-\lambda x} - \frac{1}{\lambda} (e^{-\lambda x}) \right]_0^{\infty}$$

$$= - \left[x e^{-\lambda x} \right]_0^{\infty} - \frac{1}{\lambda} \left[(e^{-\lambda x} \infty \frac{1}{e^{\lambda x}}) \right]_0^{\infty}$$

By using L. Hospital Rule.

$$x e^{-\lambda x} = \frac{x}{e^{\lambda x}} \quad x = \frac{1}{\lambda}$$

$$\lim_{x \rightarrow \infty} \frac{x}{e^{\lambda x}} = \frac{1}{\lambda e^{\lambda(\infty)}} = \frac{1}{\lambda e^{(\infty)}} = \frac{1}{\lambda(\infty)} = 0$$

or

$$\lim_{x \rightarrow 0} \frac{x}{e^{\lambda x}} = \frac{0}{e^{\lambda(0)}} = \frac{0}{e^0} = \frac{0}{1} = 0$$

$$\begin{aligned}
 E(X) &= -\frac{1}{\lambda} \left[\frac{1}{Ae^{\lambda x}} \right]_0^{\infty} \\
 &= -\frac{1}{\lambda} \left[\frac{1}{e^{\infty} - e^0} \right] = \frac{1}{\lambda} \left[\frac{1}{0 - 1} \right] = \frac{1}{\lambda} \left[\frac{1}{-1} \right] \\
 &= -\frac{1}{\lambda} \left[0 - 1 \right] \Rightarrow \frac{1}{\lambda}
 \end{aligned}$$

$$\begin{aligned}
 E(X) &= E[\text{---}] \\
 E(X) &= -0 - \frac{1}{\lambda} \left[-1 \right]
 \end{aligned}$$

A.A.D. M.C.Q.

$$E(X) = \frac{1}{\lambda}$$

Topic # 155

⇒ Derivation of the variance of exponential distrib.

As we know that

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad \lambda > 0$$

The mean is

$$E(X) = \frac{1}{\lambda}$$

As we know that

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Now

$$E(X^2) = \int_0^{\infty} x^2 f(x) dx \Rightarrow \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx \rightarrow \textcircled{1}$$

integration by parts

$$\therefore \int u dv = u \int \frac{1}{v} dx - \int \left(\frac{du}{dx} \int v dx \right) dx$$

$$u = x^2$$

$$v = \lambda e^{-\lambda x} \Rightarrow -(-\lambda e^{-\lambda x})$$

$$\int u dv = \int -(-\lambda e^{-\lambda x}) dx \Rightarrow -(e^{-\lambda x})$$

$$\frac{du}{dx} = \frac{d}{dx} (x^2) = 2x$$

$$\int x^2 \lambda e^{-\lambda x} dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$$

$$= x^2 (-e^{-\lambda x}) - \int 2x (-e^{-\lambda x}) dx$$

$$= -x^2 e^{-\lambda x} + \frac{2}{\lambda} \int x (\lambda e^{-\lambda x}) dx \Rightarrow \boxed{E(x) = \frac{1}{\lambda}}$$

$$E(x^2) = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx + \frac{2}{\lambda} E(x)$$

$$= - \left[\frac{x^2}{e^{\lambda x}} \right]_0^{\infty} + \frac{2}{\lambda} E(x) \Rightarrow -$$

let

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x^2}{e^{\lambda x}} = \lim_{x \rightarrow \infty} \frac{2x}{\lambda e^{\lambda x}} = \frac{2(0)}{\lambda^2 e^{\lambda x}}$$

$$= \frac{2}{\lambda^2 e^{\lambda(\infty)}} = \frac{2}{\infty} = 0$$

and

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2x}{e^{\lambda x}} = \frac{(0)^2}{e^{\lambda(0)}} = \frac{0}{1} = 0$$

$$= -(0-0) + \frac{2}{\lambda} \left(\frac{1}{\lambda} \right) = \frac{2}{\lambda^2}$$

$$\text{Var}(x) = E(x^2) - [E(x)]^2$$

$$= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda} \right)^2 \Rightarrow \frac{2}{\lambda^2} - \frac{1}{\lambda^2}$$

$$\text{Var}(x) = \frac{2-1}{\lambda^2} = \frac{1}{\lambda^2} \quad \text{1-M.C.P.}$$

Standard deviation:

$$\text{S.D.}(x) = \sigma = \sqrt{\frac{1}{\lambda^2}} = \frac{1}{\lambda} \quad \text{1-M.C.P.}$$

Coefficient of Variation:

$$CV = \frac{\sigma}{\mu} = \frac{\frac{1}{\lambda}}{\frac{1}{\lambda}} = \frac{1}{1} = 1 \quad \text{1-M.C.P.}$$

V.V. Imp
Topic # 156

Example: The duration of long distance telephone calls is found to be exponentially distributed with mean of 3 minutes. What is the probability that a call will last
(i) more than 3 minutes (ii) more than 5 minutes

Sol:

$$\text{let Mean} = \frac{1}{\theta}, \theta = 3$$

$$\text{Mean} = \frac{1}{3}$$

and

$$f(x) = \left(\frac{1}{\theta}\right) e^{-x/\theta} dx$$

$$\begin{aligned} \text{(i)} - P(X > 3) &= \int_3^{\infty} \left(\frac{1}{3}\right) e^{-x/3} dx = \left[-e^{-x/3} \right]_3^{\infty} \\ &= -e^{-\frac{\infty}{3}} + e^{-\frac{3}{3}} \Rightarrow 0 + e^{-1} = 0.3679 \end{aligned}$$

(ii) -

$$\begin{aligned} P(X > 5) &= \int_5^{\infty} \left(\frac{1}{3}\right) e^{-x/3} dx = \left[-e^{-x/3} \right]_5^{\infty} \\ &= -e^{-\frac{\infty}{3}} + e^{-\frac{5}{3}} \Rightarrow e^{-1.7} = 0.1827 \end{aligned}$$

Topic # 157

$\Rightarrow$ MGF of the Exponential Distribution

As we know that

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0 \end{cases}$$

where " λ " is only parameter of the distribution.

So

$$\Rightarrow M_0(t) = E(e^{tx}) = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^{\infty} e^{-\lambda x} e^{tx} dx = \lambda \int_0^{\infty} e^{-(\lambda-t)x} dx$$

$$= \lambda \left[\frac{e^{-(\lambda-t)x}}{-(\lambda-t)} \right]_0^{\infty} = \lambda \left[-\frac{e^{-(\lambda-t)x}}{(\lambda-t)} \right]_0^{\infty}$$

$$= \lambda \left[-\frac{e^{-(\lambda-t)x}}{\lambda-t} - \frac{e^{-(\lambda-t)x}}{\lambda-t} \right]_0^{\infty} = \lambda \left[-\frac{e^{-(\lambda-t)\infty}}{\lambda-t} - \frac{e^{-(\lambda-t)0}}{\lambda-t} \right]$$

$$= \lambda \left[0 - \frac{-1}{\lambda-t} \right] = \lambda \left[\frac{1}{\lambda-t} \right]$$

$$M_0(t) = \frac{\lambda}{\lambda-t} \text{ for } t < \lambda$$

Topic # 158

Gamma Distribution

The PDF of gamma distribution

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & 0 < x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

The gamma function is

$$\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy$$

Imp

Properties:

1). If $\alpha = 1$

$$\Gamma(1) = \int_0^{\infty} e^{-y} dy = 1$$

2). If $\alpha > 1$

$$\Gamma(\alpha) = (\alpha-1) \int_0^{\infty} y^{\alpha-2} e^{-y} dy$$

$$\Gamma(\alpha) = (\alpha-1) \Gamma(\alpha-1)$$

If α is positive integer greater than 1

$$\Gamma(\alpha) = (\alpha-1)(\alpha-2) \dots (3)(2)(1) \Gamma(1) = (\alpha-1)!$$

$$\Gamma(1) = 1$$

$$0! = 1$$

$$\int_0^{\infty} y^{\alpha-1} e^{-y} dy$$

$$\int_0^{\infty} y^0 e^{-y} dy = e^{-y}$$

$$= (\alpha-1)!$$

Derivation of gamma function:

$$\text{Let } y = \frac{x}{\beta} \Rightarrow x = y\beta, \frac{dx}{dy} = \beta$$

$$\frac{dx}{\beta} = dy$$

By using gamma function

$$\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy \Rightarrow \int_0^{\infty} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-x/\beta} \left(\frac{1}{\beta}\right) dx$$

$$1 = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \frac{1}{\beta^{\alpha}} x^{\alpha-1} e^{-x/\beta} dx$$

$$= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \frac{1}{\beta^{\alpha}} x^{\alpha-1} e^{-x/\beta} dx = \int_0^{\infty} \frac{1}{\Gamma(\alpha)} \cdot \frac{1}{\beta^{\alpha}} x^{\alpha-1} e^{-x/\beta} dx = 1$$

Implying that

$$\frac{1}{\Gamma(\alpha)} \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^{\alpha}}, \quad 0 < x < \infty$$

(probability density function)

2- The cdf of gamma function

$$F(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} \gamma\left(\alpha, \frac{x}{\beta}\right) & \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

⇒ The upper incomplete gamma function is

$$\Gamma(\alpha, y) = \int_y^{\infty} t^{\alpha-1} e^{-t} dt$$

⇒ The lower incomplete gamma function is

$$\gamma(\alpha, y) = \int_0^y t^{\alpha-1} e^{-t} dt$$

Note:

$$\gamma(\alpha, y) + \Gamma(\alpha, y) = \Gamma(\alpha)$$

Topic # 159

⇒ Derivation of mean and variance of gamma fun.

Mean: M.C.Q

$$\mu_x = E(x) = \alpha\beta$$

Variance: M.C.Q

$$\sigma_x^2 = \alpha\beta^2$$

⇒ Derivation of Mean:

$$\mu_x = E(x) = \alpha\beta$$

$$\int_0^{\infty} x \left(\frac{x^{\alpha-1} e^{-x/\beta}}{\beta^{\alpha} \Gamma(\alpha)} \right) dx = \int_0^{\infty} x \left[\frac{x^{\alpha-1} e^{-x/\beta}}{\beta^{\alpha} \Gamma(\alpha)} \right] dx$$

$$= \frac{\alpha}{\alpha} \int_0^{\infty} \frac{x^{\alpha} e^{-x/\beta}}{\beta^{\alpha} \Gamma(\alpha)} dx = \alpha \int_0^{\infty} \frac{x^{\alpha} e^{-x/\beta}}{\beta^{\alpha} \alpha \Gamma(\alpha)} dx = \alpha \int_0^{\infty} \frac{x^{\alpha} e^{-x/\beta}}{\beta^{\alpha} \Gamma(\alpha+1)} dx$$

$$\therefore \Gamma(\alpha+1) = \alpha \Gamma(\alpha)$$

$$= \alpha \beta \int_0^{\infty} \left(\frac{x^{\alpha} e^{-x/\beta}}{\beta^{\alpha+1} \Gamma(\alpha+1)} \right) dx = \alpha \beta (1) = \alpha \beta$$

(∴ The total area under curve of a probability M.C.Q density function is always equal to 1)

⇒ Derivation of Variance:

$$E(x^2) = \int_0^{\infty} x^2 \left(\frac{x^{\alpha-1} e^{-x/\beta}}{\beta^{\alpha} \Gamma(\alpha)} \right) dx = \alpha(\alpha+1)\beta^2$$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

$$= \alpha(\alpha+1)\beta^2 - (\alpha\beta)^2$$

$$= (\alpha^2 + \alpha)\beta^2 - \alpha^2\beta^2$$

$$= \alpha^2\beta^2 + \alpha\beta^2 - \alpha^2\beta^2$$

$$\text{Variance} = \alpha\beta^2$$

M.C.Q

Example: let x have a gamma distribution with pdf

$$f(x) = \frac{1}{\beta^2} x e^{-x/\beta}, \quad 0 < x < \infty$$

zero elsewhere, if $x=2$ is the unique mode of the distribution. find the parameter β and $P(X < 9.46)$.

Sol.

Just for check the pdf of gamma function is
 $\Rightarrow f(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$

Put $\alpha = 2$

$$= \frac{1}{\Gamma(2)\beta^2} x^{2-1} e^{-x/\beta} = \frac{1}{1!\beta^2} x e^{-x/\beta}$$

$$= \frac{1}{\beta^2} x e^{-x/\beta}, \quad 0 < x < \beta$$

$\Rightarrow$ As we know that

$$\Rightarrow \text{Mode} = (\alpha - 1)\beta$$

Put $\alpha = 2$ and Mode = 2

$$\text{Mode} = (2 - 1)\beta \Rightarrow \beta$$

$$2 = (2 - 1)\beta \Rightarrow \beta = \frac{2}{1} = 2$$

Now

$$f(x) = \frac{1}{\beta^2} x e^{-x/\beta} \Rightarrow \frac{1}{2^2} x e^{-x/2} dx$$

$$= \frac{1}{4} x e^{-x/2} dx \Rightarrow \int_0^{9.46} \frac{1}{4} x e^{-x/2} dx$$

$$= \frac{1}{4} \int_0^{9.46} x e^{-x/2} dx$$

$$= \frac{1}{4} \left[-2 e^{-x/2} x - 4 e^{-x/2} \right]_0^{9.46}$$

$$= \frac{1}{4} \left[(2 e^{-9.46/2} - 4 e^{-9.46/2}) - (2 e^{-0/2} - 4 e^{-0/2}) \right]$$

$$= \frac{1}{4} [(0.16700 - 0.03531) - (-0 - 4e^0)]$$

$$= \frac{1}{4} [(-0.20230) - (-4)] = \frac{1}{4} [-0.20230 + 4]$$

$$= \frac{3.79770}{4} = 0.9494$$

$P(X < 9.46)$ is 94.94% almost 95%.

Topic # 161

→ MGF of gamma distribution

As

$$M(t) = (1 - \beta t)^{-\alpha} \quad \text{for } t < \frac{1}{\beta}$$

M.C.P.

Topic # 162

PDF M.C.P. Beta distribution

→ the probability density function of beta distribution is

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} \quad 0 \leq x \leq 1, \alpha, \beta > 0$$

M.C.P.

where

$$B(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

Beta function plays to role the denominator and the area under the curve is 1

$$\Gamma(k) = (k-1)!$$

Cumulative distribution function

$$F(x) = I_x(\alpha, \beta) = \frac{\int_0^x t^{\alpha-1} (1-t)^{\beta-1} dt}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1; \alpha, \beta > 0$$

M.C.P.

Most properties: M.C.P.

1) Mean = $\mu \Rightarrow E(X) = \frac{\alpha}{\alpha + \beta}$

$$f_{\alpha}(x) = \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)\Gamma(1)} x^{\alpha} (1-x)^{0} = x^{\alpha} \quad \text{if } x \in [0,1]$$

$$= 0 \quad \text{if } x \notin [0,1]$$

M.C.Q

2-) Variance (x) = $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Mode :

$$\text{Mode} = \frac{\alpha-1}{\alpha+\beta-2} \quad \text{M.C.Q}$$

Topic # 163

Derivation of the Mean and Variance of the Beta distribution.

Mean:

$$\mu = E(X) = \frac{\alpha}{\alpha+\beta} \quad \left| \quad E(X^2) = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} \right.$$

Variance:

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ \sigma^2 &= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta} \right)^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} \end{aligned}$$

Topic # 164

Important Note

⇒ There are two-type of beta distribution

- 1)- beta distribution of the first kind is defined on the interval from 0 to 1
- 2)- beta distribution of the second kind is defined on the interval from 0 to infinity.

Example: Suppose that the proportion of defective DVDs in a large shipment follow a Beta distribution with $\alpha=2$ and $\beta=5$. What is the probability that the shipment has 20% to 30% defective DVDs?

Sol.

As we know that

$$f(x) = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)} \quad 0 \leq x \leq 1, \alpha > 0, \beta > 0$$

put $\alpha=2$ and $\beta=5$

$$f(x) = \frac{x^{2-1} (1-x)^{5-1}}{B(2, 5)}, \quad 0 \leq x \leq 1$$

$$= \frac{x(1-x)^4}{B(2, 5)} \Rightarrow$$

By given condition

$$P(20\% \leq x \leq 30\%) = P(0.20 \leq x \leq 0.30)$$

and

$$P(0.20 \leq x \leq 0.30) = \int_{0.2}^{0.3} \frac{x^{2-1} (1-x)^{5-1}}{B(2, 5)} dx$$

for beta function

$$B(a, b) = \frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)} \Rightarrow B(2, 5) = \frac{\Gamma(2) \Gamma(5)}{\Gamma(2+5)}$$

$$= \frac{2! 5!}{7!} \quad \therefore K! = (K-1)!$$

$$= \frac{1! 4!}{6!} = 0.0333$$

Now

$$P(0.2 \leq x \leq 0.3) = \int_{0.2}^{0.3} \frac{x^{2-1} (1-x)^{6-1}}{0.0333} dx$$

$$= \frac{1}{0.0333} \int_{0.2}^{0.3} x (1-x)^5 dx$$

integrating by parts
 $P(0.2 \leq x \leq 0.3) = 0.2352$

Topic # 165 V.V. Imp Normal distribution

$\Rightarrow$ Definition: A random variable X is said to be normal distribution

$\Rightarrow$ If its (PDF) is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \text{ for } -\infty < x < \infty$$

1) μ is mean its median and mode

2) $\sigma > 0$ is the standard deviation

3) σ^2 is the variance

$$\text{Mode} = \frac{1}{\sigma\sqrt{2\pi}}$$

$\Rightarrow$ Cumulative distribution function is

$$F(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x-\mu}{\sigma\sqrt{2}} \right) \right] \text{ for } -\infty < x < \infty$$

$\Rightarrow$ Error function:

M.C.Q

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$$

Note: ^{standard normal distribution} It is the distribution with a normal random variable a mean of zero and a standard deviation of one.

Topic # 166

Standard Normal Distribution

If the random variable Z follows a normal distribution with parameters $(0,1)$

i.e. $Z \sim N(0,1)$ is called standard normal random variable.

$\Rightarrow$ The Pdf of standard normal distribution

Let

$$f_z(z) = \frac{1}{\sqrt{2\pi}} e^{\left(\frac{-z^2}{2}\right)} \quad -\infty < z < \infty$$

M.C.O

$\Rightarrow$ For odd-function,

$$z f_z(z) = z \cdot \frac{1}{\sqrt{2\pi}} e^{\left(\frac{-z^2}{2}\right)}$$

$$z f_z(z) = \frac{z}{\sqrt{2\pi}} e^{\left(\frac{-z^2}{2}\right)} \quad \text{M.C.O}$$

$\Rightarrow$ Mean: M.C.O

$$E(Z) = \int_{-\infty}^{\infty} z f_z(z) dz = 0$$

$\Rightarrow$ Variance:

$$\text{Var}(Z) = E(Z^2) - [E(Z)]^2$$

$$\text{put } E(Z) = 0$$

$$= E(Z^2) - 0 \Rightarrow E(Z^2)$$

$$= \int_{-\infty}^{\infty} z^2 \frac{e^{\left(\frac{-z^2}{2}\right)}}{\sqrt{2\pi}} dz$$

$\Rightarrow$ For even-function

$$= z^2 \frac{e^{\left(\frac{-z^2}{2}\right)}}{\sqrt{2\pi}} \text{ is even}$$

Odd function
A function for which
 $g(-x) = -g(x) \quad \forall x \in \mathbb{R}$
 $\int_{-\infty}^{\infty} g(x) dx = 0$
 $\int_{-\infty}^{\infty} g(x) dx = 0$

Even function
A function for which
 $g(-x) = g(x) \quad \forall x \in \mathbb{R}$
 $\int_{-\infty}^{\infty} g(x) dx = 2 \int_0^{\infty} g(x) dx$

$$\text{Var}(Z) = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z^2 e^{-\frac{z^2}{2}} dz$$

let

$$\text{Var}(Z) = 1 \quad \text{M.C.D.}$$

$$\begin{aligned} w &= \frac{z^2}{2} \\ dw &= z dz \\ \sqrt{2}z &= \sqrt{dw} \\ z &= \frac{\sqrt{dw}}{\sqrt{2}} \\ z &= \frac{1}{\sqrt{2}} \sqrt{dw} \end{aligned}$$

Topic # 167

$$Z = \frac{X - \mu}{\sigma} \quad \text{Formula}$$

⇒ The Process of standardization:

Let X be $N(\mu, \sigma^2)$ and suppose that we are interested to computing the probability $P(a < X < b)$.

and $N(\mu, \sigma^2)$ random variable X into the standard normal random variable $Z \sim N(0, 1)$.

V.V. Imp

Example: Let X be $N(2, 25)$ and suppose that we are interested in computing the probability $P(0 < X < 10) = F(10) - F(0)$.

Sol. let $\mu = 2$, $\sigma^2 = 25 \Rightarrow \sigma = 5$
AS applying standardization formula

$$= \frac{X - \mu}{\sigma} = \frac{X - 2}{5}$$

Now

$$P(0 < X < 10) = F(10) - F(0)$$

$$= \Phi\left(\frac{10-2}{5}\right) - \Phi\left(\frac{0-2}{5}\right)$$

$$= \Phi\left(\frac{8}{5}\right) - \Phi\left(-\frac{2}{5}\right) \rightarrow \Phi(1.60) - \Phi(-0.40)$$

$$= 0.945 - (1 - 0.655) \\ = 0.600 \Rightarrow 60\%$$

Cdf of standard normal distribution

put

$$X=10, X=0$$

using

Area using table values

$$z = \frac{X - \mu}{\sigma} \Rightarrow z\sigma = X - \mu \Rightarrow X = \mu + z\sigma$$

Topic # 168 Body Mass Index for Percentiles

M.C.Q.

=> Computation of Some Specific Percentiles

Example: The BMI for Men aged 60 is normally distributed with mean equal to 29 with a standard deviation of 6 whereas the BMI for Women aged 60 is normally distributed with mean 28 and (s.d.) 7.

=> What is the 90th percentile of BMI for men and what does it mean?

Sol.

To compute the 90th percentile we use the formula

$$X = \mu + z\sigma$$

By using area table we get 0.8997 is much closer to 0.9015

The z-score is 1.28

Let $\mu = 29$, $\sigma = 6$

$$X = 29 + 6(1.28)$$

$$= 29 + 7.68 \Rightarrow 36.68$$

The 90th percentile is 36.68

=> The Ninety percent of the BMI is men aged 60 are 36.68 or less

=> Ten percent of the BMI is men aged 60 are 36.68 or higher

$$\begin{array}{r|l} 90\% & \\ \hline 0.90 & \\ 0.9000 & \end{array}$$

$$\begin{array}{l} \text{1st column} = z = 1.2 \\ \text{Top row} = z = 0.08 \\ z = 1.28 \end{array}$$

M.C.Q.
At 90th percent
 $z = 1.282$

Topic # 169

The total area under the curve.

$$\text{Area} = \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Let

$$x = \mu + z\sigma \Rightarrow \frac{x}{\sigma} = \mu + z \Rightarrow \frac{dx}{dz} = 0 + \sigma \Rightarrow dx = \sigma dz$$

let

$$v = \frac{1}{2} z^2 \Rightarrow \frac{dv}{dz} = \frac{1}{2} \cdot 2z = z \Rightarrow dv = z dz$$

$$dz = \frac{dv}{\sqrt{v}}$$

$$\text{Area} = \frac{1}{\sigma\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1$$

$$\text{Area} = 1$$

$\Rightarrow$ (The total area (Probability) under the normal curve is unity.) M.C.Q

Topic # 170

$\Rightarrow$ Derivation of the Mean and a Variance of the normal distribution.

M.C.Q

Mean:

$$E(X) = \mu$$

M.C.Q

Variance:

$$\text{Var}(X) = \sigma^2$$

Topic # 171

$\Rightarrow$ The Median and the Mode of the Normal distribution.

Median:

$$\text{Let } m \int_{-\infty}^m f(x) dx = \frac{1}{2}$$

m - the median of the distribution

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^m e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{2}$$

$$\text{Median} = \frac{1}{2}$$

$$\text{Mode: } f'(x) = 0$$

$$\text{and } f''(x) < 0$$

$$\frac{m-\mu}{\sigma} = 0$$

$$m-\mu = 0$$

$$m = \mu$$

μ is median of data

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx$$

$$f'(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \cdot \frac{-(x-\mu)}{\sigma^2} \cdot x$$

$$= -\frac{1}{\sigma^3\sqrt{2\pi}} (x-\mu) e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

and

$$f''(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} \left[e^{-\frac{(x-\mu)^2}{2\sigma^2}} - \frac{(x-\mu)^2}{\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right]$$

$$f''(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \left[1 - \frac{(x-\mu)^2}{\sigma^2} \right]$$

Since

mean = median = mode

the normal distribution is symmetrical and unimodal.

Topic # 172

⇒ The Quartile Deviation of the Normal distribution is approximately $\left(\frac{2}{3}\right)^{\text{rd}}$ of its standard deviation.

The Semi-Inter-Quartile Range or Quartile deviation is

$$Q = \frac{Q_3 - Q_1}{2} \quad \text{M.C.Q.}$$

For a Symmetric.

$$Q = Q_2 - Q_1 \quad \text{and} \quad Q = Q_3 - Q_2$$

$$\text{let } Q_2 = \mu$$

$$Q = \mu - Q_1 \quad , \quad \text{and} \quad Q = Q_3 - \mu$$

Topic # 173

⇒ The ^{imp def.} normal curve has Point of Inflection.

^{M.C.Q.} Definition: By a "Point of Inflection" we mean a point at which the concavity of a curve changes.

Note: ^{M.C.Q.}

⇒ The point of inflection on the right and on the left of the mean at a distance equal to Standard deviation

⇒ ^{M.C.Q.} The graph of the normal curve is (bell-shaped)

Hence, the two point of inflection of normal curve are

$$\left(\mu - \sigma, \frac{1}{\sigma \sqrt{2\pi}} \right) \quad \text{and} \quad \left(\mu + \sigma, \frac{1}{\sigma \sqrt{2\pi}} \right)$$

Topic # 174

Theorem: For the normal distribution, the odd order moments about the mean are all zero and the even order moments are given by

$$\mu_{2n} = (2n-1)(2n-3) \dots 5 \cdot 3 \cdot 1 \sigma^{2n}$$

Proof — The odd order moments about the mean is

$$\mu_{2n+1} = E(X-\mu)^{2n+1} = \int_{-\infty}^{\infty} (x-\mu)^{2n+1} \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

where $n \in \mathbb{N}$

Substituting $z = \frac{x-\mu}{\sigma} \Rightarrow \sigma z = x - \mu$

$$x = \mu + \sigma z, \quad \frac{dx}{dz} = \sigma \Rightarrow dx = \sigma dz$$

$$\Rightarrow \mu_{2n+1} = \int_{-\infty}^{\infty} (\sigma z)^{2n+1} \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{1}{2}z^2} \sigma dz$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} (\sigma z)^{2n+1} e^{-\frac{1}{2}z^2} dz$$

$$= \frac{\sigma^{2n+1}}{\sqrt{\pi}} \int_{-\infty}^{\infty} z^{2n+1} e^{-\frac{1}{2}z^2} dz = 0$$

$\Rightarrow z^{2n+1} e^{-\frac{1}{2}z^2}$ is odd function of z

$\Rightarrow$ Thus $\mu_1 = \mu_3 = \mu_5 = \dots = 0$

Topic # 175

⇒ Derivation of the Variance of the Exponential Distribution.

Note: ^{M.C.Q} The Mean and Standard deviation of the exponential distribution are equal.

⇒ **Mean:**

let $f(x) = \lambda e^{-\lambda x}$, $x \geq 0, \lambda > 0$
so

$$E(x) = \mu = \frac{1}{\lambda}$$

and

$$E(x^2) = \frac{2}{\lambda^2}$$

⇒ ^{M.C.Q} $\text{Var}(x) = E(x^2) - [E(x)]^2$

$$= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} \Rightarrow \frac{2-1}{\lambda^2} = \frac{1}{\lambda^2}$$

^{M.C.Q} $\text{Var}(x) = \frac{1}{\lambda^2}$

⇒ ^{M.C.Q} Standard deviation = $\sigma = \frac{1}{\lambda}$

⇒ ^{M.C.Q} Coefficient of Variation = $CV = \frac{\sigma}{\mu}$

$$CV = \frac{\sigma}{\mu} = \frac{\frac{1}{\lambda}}{\frac{1}{\lambda}}$$

$$CV = \frac{1}{\lambda} \times \lambda \Rightarrow 1$$

^{M.C.Q} **Note:** The coefficient of Variation of the exponential distribution is always equal to 1.

Topic # 176

⇒ obtaining the first two Raw Moments of the Normal distribution from its MGF.
we know that, the mgf of normal distribution is

$$M(t) = e^{\mu t + \frac{1}{2} \sigma^2 t^2}, \text{ for } -\infty < t < \infty$$

⇒ For 1st derivative: (first Raw Moment)

$$M'_x(t) = \frac{d}{dt} \left[e^{\mu t + \frac{1}{2} \sigma^2 t^2} \right]$$

$$= e^{\mu t + \frac{1}{2} \sigma^2 t^2} \cdot \frac{d}{dt} \left[\mu t + \frac{1}{2} \sigma^2 t^2 \right]$$

$$= e^{\mu t + \frac{1}{2} \sigma^2 t^2} \cdot \left[\mu + \frac{1}{2} \sigma^2 \cdot 2t \right] \Rightarrow \left[\mu + \sigma^2 t \right]$$

put $t=0$

$$M'_x(0) = e^{\mu(0) + \frac{1}{2} \sigma^2 (0)^2} \cdot [\mu + \sigma^2(0)]$$

$$M'_x(0) = e^{0+0} [\mu + 0] \Rightarrow e^0 [\mu + 0] \Rightarrow 1 \cdot \mu$$

$$M'_x(0) = \mu$$

$$E(x) = \mu$$

⇒ Similarly for Second derivative

$$E(x^2) = ? \text{ (second Raw Moment)}$$

$$M''_x(0) = \mu^2 + \sigma^2$$

$$\text{so, } E(x^2) = \mu^2 + \sigma^2$$

Variance = ?

$$\Rightarrow \text{Var}(x) = E(x^2) - [E(x)]^2$$

$$= \mu^2 + \sigma^2 - \mu^2$$

$$\text{Var}(x) = \sigma^2$$

Topic # 177

⇒ Cumulant Generating Function of the ^{normal} $N(\mu, \sigma^2)$ distribution and its utilization for finding its cumulants.

As we know that the cumulant generating function $K(t)$ is the natural logarithm of the moment generating function ^{M.G.F.}

$$K(t) = \ln E(e^{tx})$$

⇒ For the normal distribution: (CGF)

^{proof} $K(t) = \mu t + \frac{\sigma^2 t^2}{2}$ ^{M.G.F.}

* For cumulants

⇒ we find the first derivative (1st derivative)

$$K'(t) = \frac{d}{dt} \left(\mu t + \frac{\sigma^2 t^2}{2} \right)$$

$$= \mu \frac{d}{dt} (t) + \frac{d}{dt} \left(\frac{\sigma^2 t^2}{2} \right)$$

$$K'(t) = \mu(1) + \frac{\sigma^2}{2} (2t) \Rightarrow \mu + \sigma^2 t$$

⇒ For second derivative:

$$K''(t) = \frac{d}{dt} (K'(t)) \Rightarrow \frac{d}{dt} (\mu + \sigma^2 t)$$

$$= \frac{d}{dt} (\mu) + \sigma^2 \frac{d}{dt} (t)$$

$$K''(t) = 0 + \sigma^2 (1) \Rightarrow \sigma^2$$

⇒ For third derivative:

$$K'''(t) = \frac{d}{dt} (K''(t)) \Rightarrow \frac{d}{dt} (\sigma^2)$$

$$K'''(t) = 0$$

1) For First Cumulant:

$$\text{put } t=0$$

$$K'(0) = (\mu + \sigma^2(0))_{t=0} = \mu$$

$$K'(0) = \mu$$

2) For Second Cumulant:

$$\text{put } t=0$$

$$K''(0) = (\sigma^2)_{t=0} = \sigma^2$$

$$K''(0) = \sigma^2$$

3) For Third Cumulant:

$$\text{put } t=0$$

$$K'''(0) = (0)_{t=0} = 0$$

Thus

$$K_1 = \mu, \quad K_2 = \sigma^2$$

and

$$K_3 = K_4 = \dots = 0$$

17.10.20

For Thm's

of both

$np \geq 5$

and

$np \geq 5$

Topic # 17.8

Example: A fair coin is tossed 20 times find the Probability That the number of heads occurring is b/w 10 and 14 inclusive by using

- The binomial distribution.
- The normal approximation to the binomial distribution.

Sol. let us consider the pmf of X is

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x} \quad \text{or } \binom{n}{x} \left(\frac{1}{2}\right)^n \quad \text{fair coin}$$

$$\text{put } n=20, \quad p=\frac{1}{2}$$

$$= \binom{20}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{20-x}$$

$$= \binom{20}{x} \left(\frac{1}{2}\right)^{20-x+x} \rightarrow \binom{20}{x} \left(\frac{1}{2}\right)^{20}$$

put $x = 10, 11, 12, 13$ and 14

$$P(10 \leq X \leq 14) = \sum_{x=10}^{14} \binom{20}{x} \left(\frac{1}{2}\right)^{20}$$

$$P(10 \leq X \leq 14) = \binom{20}{10} \left(\frac{1}{2}\right)^{20} + \binom{20}{11} \left(\frac{1}{2}\right)^{20} +$$

$$\binom{20}{12} \left(\frac{1}{2}\right)^{20} + \binom{20}{13} \left(\frac{1}{2}\right)^{20} + \binom{20}{14} \left(\frac{1}{2}\right)^{20}$$

$$= \frac{20!}{10!(20-10)!} \left(\frac{1}{2}\right)^{20} + \frac{20!}{11!(20-11)!} \left(\frac{1}{2}\right)^{20} + \frac{20!}{12!(20-12)!} \left(\frac{1}{2}\right)^{20}$$

$$+ \frac{20!}{13!(20-13)!} \left(\frac{1}{2}\right)^{20} + \frac{20!}{14!(20-14)!} \left(\frac{1}{2}\right)^{20}$$

$$= 0.1762 + 0.1602 + 0.1201 + 0.0739 + 0.0370$$

$$P(10 \leq X \leq 14) = 0.5674$$

b) - For normal approximation to Binomial dist.

$$\text{Here } n=20, \quad p=\frac{1}{2}, \quad q=\frac{1}{2}$$

$$\text{So } np = 20 \times \frac{1}{2} = 10 > 5$$

$$nq = 20 \times \frac{1}{2} = 10 > 5$$

$\Rightarrow$ For normal distribution:

$$\mu = np = (20)(0.5) = 10$$

and

$$\sigma = \sqrt{npq} = \sqrt{(20)(0.5)(0.5)} = 2.24$$

$\Rightarrow$ For Area:

we compute z-values

$$X = 10, 11, 12, 13 \text{ and } 14$$

$$\text{let } x = 9.5$$

$$z = \frac{x - \mu}{\sigma} = \frac{9.5 - 10}{2.24} = -0.22$$

$$\text{let } x = 14.5$$

$$z = \frac{14.5 - 10}{2.24} = 2.01$$

By using table

$$P(9.5 \leq X \leq 14.5) = P(-0.22 \leq Z \leq 2.01)$$

So,

$$= 0.0871 + 0.4778 = \boxed{0.5649} \text{ Ans}$$

~~V.V. Imp~~

Topic # 179

Normal Approximation to Poisson distribution

Example: The number of calls received by an office switch board per hour follows a Poisson distribution with parameter 25.

$\Rightarrow$ Find the Probabilities that in one hour

(a) There are b/w 23 and 26 calls

(b) - more than 30 calls, using the normal approximation to the Poisson distribution.

(9) -
Sol.

Let X is $P(x, 25)$
we require

a) $P(23 \leq x \leq 26)$ and (b) $P(x > 30)$

and the normal X is $N(25, 25)$ $\sigma^2 = \sqrt{25}$
 $\sigma = 5$

$\Rightarrow$ The z -values are:

put $x = 23, 24, 25$, and 26 .

put $x = 22.5$

$$z = \frac{x - \mu}{\sigma} = \frac{22.5 - 25}{5} = -0.5$$

at

$$x = 26.5$$

$$z = \frac{26.5 - 25}{5} = 0.3$$

So,

$$P(22.5 \leq x \leq 26.5) = P(-0.5 \leq z \leq 0.3)$$

$$\Rightarrow \text{using Area table} = 0.1915 + 0.1179$$

$$= 0.3094 \times 100\%$$

$$= 30.94\%$$

$$\text{b) - } P(x \geq 30) \text{ or } P(x > 29.5):$$

$$\mu = 25 \text{ or } \lambda = 25$$

$$P = \left(\frac{x - 29.5}{5} > \frac{29.5 - 25}{5} \right)$$

$$P = \left(z > \frac{4.5}{5} \right) \Rightarrow P(z > 0.9)$$

$$= 0.1841 \Rightarrow 0.1841 \times 100$$

$$= 18.41\%$$

- (i) outer circle - very short
 (ii) ordinates against second circle a little taller
 (iii) ordinates against 3rd circle a taller than one and second
 (iv) similarly fourth -

Topic # 180

⇒ Bivariate Normal distribution in PDF

Defination: A pair of random variables X_1 and X_2 have a bivariate normal distribution and they are referred to as jointly normally distributed random variables if and only if their joint probability density is given by

$$f(x_1, x_2) = \frac{1}{\sigma_1 \sigma_2 \sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left(\frac{(x_1 - \mu_1)^2}{\sigma_1^2} - 2\rho \frac{(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_1 \sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right)}$$

for $-\infty < x_1 < \infty$ and $-\infty < x_2 < \infty$

n.c.d
 Joint P.d.f
 well-shaped

Now,

$$N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$$

Contour:

Set of Points on the x_1, x_2 Plane

corresponding ordinates are equal height

A few real-life examples (Bivariate dist)

- (i) - Heights and weights
- (ii) - weights and Blood pressure
- (iii) - Marks in Mathematics and Marks in statistics.

Eq. of ellipse:

$$\frac{(x_1 - c_1)^2}{a^2} + \frac{(x_2 - c_2)^2}{b^2} = 1$$

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For centre (c_1, c_2) with radii a

$$(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

$$\text{or } (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 = 100$$

Topic # 182

Example: A statistics class takes two exams. Exam 1 (Midterm Exam) and Exam 2 (Final Exam). and let us suppose that the marks of the two exams (to be called X and Y) follow a bivariate normal distribution with parameters $\mu_x = 70$, $\mu_y = 60$ (which called marginal means) $\sigma_x^2 = 10$ and $\sigma_y^2 = 15$ (which called marginal s.d.) and $\rho = 0.6$

- what is the probability that the students scores over 75 on Exam 1 (mid term)
- what is the probability that the students scores over 85 on Exam 2 (final term) given that she/he scores 75 in mid term.

Sol. (a) - let

Mean = 70 and Variance = 100

$$\text{So } X \sim N(70, 100)$$

the probability

$$P(X > 75) = P\left(Z > \frac{X - \mu_x}{\sigma_x}\right) = P\left(Z > \frac{75 - 70}{10}\right)$$

$$= P(Z > 0.5)$$

By using area table = 0.3085 x 100

Now $x = 75$ = 30.85%

$$f_{Y|X}(y|x=75) = \frac{1}{\sqrt{2\pi}\sigma_y\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(y - \left[\mu_y + \rho\frac{\sigma_y}{\sigma_x}(x - \mu_x)\right]\right)^2}$$

variance
 $\sigma_x^2 = 10$
 $\sigma_x = \sqrt{10}$
 $\sigma_x^2 = 100$

$$\begin{aligned}
 &= \frac{1}{\sqrt{2\pi}(15)\sqrt{1-(0.6)^2}} e^{-\frac{1}{2(15)^2(1-0.36)} \left(y - [60 + (0.6)(15)] \right)^2} \\
 &= \frac{1}{\sqrt{2\pi}(15)(1-0.36)} e^{-\frac{1}{2(12)^2} \left(y - [60 + (0.6)(75-60)] \right)^2} \\
 &= \frac{1}{(12)\sqrt{2\pi}} e^{-\frac{1}{2(12)^2} (y - 64.5)^2} \\
 &= \frac{1}{(12)\sqrt{2\pi}} e^{-\frac{1}{2(12)^2} (y - 64.5)^2}
 \end{aligned}$$

(ii) $\mu_y = 64.5$ $\sigma_y = 12$

$$\begin{aligned}
 P(Y > 85) &= P\left(Z > \frac{Y - \mu_y}{\sigma_y}\right) = P\left(Z > \frac{85 - 64.5}{12}\right) \\
 &= P(Z > 1.71) \\
 &= 0.0436 \Rightarrow 0.0436 \times 100 \% \\
 &= 4.36 \%
 \end{aligned}$$

Topic # 183

$\Rightarrow$ MGF of Bivariate Normal distribution

Let

$$M_x(t) = M_x(t_1, t_2) = E\left(e^{\sum_{i=1}^2 t_i x_i}\right)$$

and

$$\begin{aligned}
 &\mu_1 t_1 + \mu_2 t_2 + \frac{1}{2} \left(\sigma_1^2 t_1^2 + 2\rho\sigma_1\sigma_2 t_1 t_2 + \sigma_2^2 t_2^2 \right) \\
 &= e^{-\frac{1}{2} \left(\sigma_1^2 t_1^2 + 2\rho\sigma_1\sigma_2 t_1 t_2 + \sigma_2^2 t_2^2 \right)}
 \end{aligned}$$

For positive constant

$$E(x_1) = \frac{\partial M(0,0)}{\partial t_1}, \quad E(x_2) = \frac{\partial M(0,0)}{\partial t_2}$$

$$E(x_1)^2 = \frac{\partial^2 M(0,0)}{\partial t_1^2}, \quad E(x_2)^2 = \frac{\partial^2 M(0,0)}{\partial t_2^2}$$

$$E(x_1 x_2) = \frac{\partial^2 M(0,0)}{\partial t_1 \partial t_2}$$

$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2}$, $-\infty < x < \infty$
 $E(X) = \mu_X = 11x$, $Var(X) = \sigma_X^2$
 $E(Y) = \mu_Y$, $Var(Y) = \sigma_Y^2$

Topic # 185

⇒ Definition of independence:

Two random variables 'X' and 'Y' are independent,

Statistically independent, Stochastically independent

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Theorem: If X and Y have a bivariate normal distribution with correlation coefficient ρ . Then X and Y are independent if and only if $\rho = 0$

1)- If X and Y are independent then $\rho = 0$

2)- If $\rho = 0$ then X and Y are independent

Sol.

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]}$$

put $\rho = 0$

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-0^2}} e^{-\frac{1}{2(1-0)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 + 2(0) \right]}$$

$$= \frac{1}{2\pi\sigma_X\sigma_Y} e^{-\frac{1}{2} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right]}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2} \times \frac{1}{\sqrt{2\pi}\sigma_Y} e^{-\frac{1}{2}\left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}$$

$$f_{XY}(x, y) = f_X(x) \times f_Y(y)$$

STA642

Probability Distribution

Short Notes for Final Term

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Note:

PMF; Probability Mass Function

PDF; Probability Density Function

CDF; Cumulative Distribution Function

MGF; Moment Generating Function

CGF; Cumulant Generating Function

Formal Definition of Independence in the case of Discrete Random Variables

Let the random variables X_1 and X_2 have the joint pmf $p(x_1, x_2)$ and the marginal pmfs $p_1(x_1)$ and $p_2(x_2)$ respectively.

The random variables X_1 and X_2 are said to be independent **if and only if**

$$p(x_1, x_2) = p_1(x_1) p_2(x_2).$$

Theorem:

If X and Y are random variables with joint distribution function F and marginal distribution functions F_X and F_Y , then X and Y are independent if and only if

$$F(x, y) \equiv F_X(x) F_Y(y)$$

For all $(x, y) \in \mathbb{R}^2$.

Proof:

If X and Y are independent, then

$$F(x, y) = P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) = F_X(x)F_Y(y).$$

Theorem:

Suppose X_1 and X_2 are independent and that $E(u(X_1))$ and $E(v(X_2))$ exist.

Then

$$E[u(X_1)v(X_2)] = E[u(X_1)]E[v(X_2)].$$

Theorem:

Suppose the joint mgf, $M(t_1, t_2)$, exists for the random variables X_1 and X_2 .

Then, X_1 and X_2 are independent if and only if

$$M(t_1, t_2) = M(t_1, 0)M(0, t_2);$$

That is, the joint mgf is identically equal to the product of the marginal mgfs.

PMF of the Discrete Uniform Distribution

If x is assuming values $1, 2, 3, \dots, n$, then the probability mass function (PMF) of the discrete uniform distribution is given by

$$p(x) = \begin{cases} \frac{1}{n}, & x = 1, 2, 3, \dots, n \\ 0 & \text{elsewhere} \end{cases}$$

CDF of the Discrete Uniform Distribution

$$p(X = x) = \begin{cases} \frac{1}{b - a + 1}, & x = a, a + 1, a + 2, \dots, b - 1, b \\ 0 & \text{elsewhere} \end{cases}$$

Mean of the Discrete Uniform Distribution

$$\mu = E(X) = \frac{n+1}{2}$$

Variance of the Discrete Uniform Distribution

$$\sigma = E(X^2) - (E(X))^2 = \sqrt{\frac{n^2 - 1}{12}}$$

MGF of the Discrete Uniform distribution

Moment Generating Function:

$$M(t) = E(e^{tx}) = \sum_{x=1}^N \left(e^{tx} \frac{1}{n} \right) = \frac{1}{n} \sum_{x=1}^n (e^t)^x = \frac{e^t - e^{(n+1)t}}{n(1 - e^t)}$$

PMF of the Binomial distribution

Let the random variable X denote the number of successes in n trials --- when a binomial experiment is performed.

Then the PMF of X is given by

$$p(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x = 0, 1, 2, \dots, n \\ 0 & \text{elsewhere.} \end{cases}$$

where p is the probability of success in each trial and n is the number of trials.

Binomial Theorem

$$(q + p)^n = \sum_{x=0}^n \binom{n}{x} p^x q^{n-x}$$

Mean of the Binomial Theorem

$$\mu = np$$

Variance of the Binomial Theorem

$$\sigma^2 = np(1-p) = npq$$

MGF of the Binomial distribution

The mgf of a binomial distribution is easily obtained as follows:

$$\begin{aligned} M(t) &= \sum_x e^{tx} p(x) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (e^t p)^x (1-p)^{n-x} \\ &= [(1-p) + pe^t]^n \\ M(t) &= (1-p + pe^t)^n \end{aligned}$$

for all real values of t .

So, in case of m binomial random variables X_i , $i = 1, 2, \dots, m$,

with $p_1 = p_2 = p_3 \dots p_m = p$ (i.e. a constant)

the mgf of X_i is $M_{X_i}(t) = (1 - p + pe^t)^{n_i}$

PMF of the Negative Binomial Distribution

In a sequence of such trials, let the random variable X denote the number of the trial at which the r^{th} success occurs, where r is a fixed integer.

$$P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, r+2, \dots$$

and we say that X has a negative binomial (r, p) distribution.

Mean of the Negative Binomial Distribution

$$E(X) = \mu = \frac{r(1-p)}{p} = \frac{rq}{p}$$

Variance of the Negative Binomial Distribution

$$\text{Var}(X) = \sigma^2 = \frac{r(1-p)}{p^2} = \frac{rq}{p^2}$$

MGF of the Negative Binomial Distribution

$$M(t) = \left(\frac{1-p}{1-pe^t} \right)^r \quad \text{for } t < -\log p$$

A Special Case of the Negative Binomial Distribution: Geometric Distribution

In a sequence of independent Bernoulli (p) trials, let the random variable X denote the number of the trial at which the 1^{st} success occurs.

PMF of the Geometric Distribution

$$\begin{aligned} P(X = x) &= \binom{x-1}{0} p^1 (1-p)^{x-1} \\ &= (1-p)^{x-1} p, \quad x = 1, 2, 3, \dots \end{aligned}$$

Mean of the Geometric Distribution

$$\mu = E(X) = \frac{1}{p}$$

Variance of the Geometric Distribution

$$\sigma^2 = \text{Var}(X) = \frac{1-p}{p^2} = \frac{q}{p^2}$$

Multinomial Distribution

The binomial distribution is generalized to be the multinomial distribution as follows:

Let p_i be the probability that the outcomes is an element of C_i and let p_i remain constant throughout the n independent repetitions, $i = 1, 2, \dots, k$.

PMF of the Multinomial Distribution

$$f(x_1, x_2, \dots, x_k; n, p_1, p_2, \dots, p_k) = \begin{cases} \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} \times p_2^{x_2} \dots \times p_k^{x_k}, & \text{when } \sum_{i=1}^k x_i = n \\ 0 & \text{otherwise} \end{cases}$$

for non-negative integers $x_1, x_2, \dots, x_k$.

Mean of the Multinomial Distribution

$$\mu = E(X_i) = np_i$$

Variance of the Multinomial Distribution

$$\sigma^2 = \text{Var}(X_i) = np_i(1 - p_i) = np_i q_i$$

MGF of the Multinomial Distribution

$$M_{\underline{X}}(t) = M_{\underline{X}}(t_1, t_2, \dots, t_k) = E\left(e^{\sum_{i=1}^k t_i X_i}\right) = \left(\sum_{i=1}^k p_i e^{t_i}\right)^n$$

CGF of the Multinomial Distribution

Cumulant Generating Function:

$$K_{\underline{X}}(t) = K_{\underline{X}}(t_1, t_2, \dots, t_k) = \log[M_{\underline{X}}(t_1, t_2, \dots, t_k)] = n \log\left(\sum_{i=1}^k p_i e^{t_i}\right)$$

MGF of the Trinomial Distribution

$$M_{\underline{X}}(t) = M_{X_1, X_2, X_3}(t_1, t_2, t_3) = E\left(e^{\sum_{i=1}^3 t_i X_i}\right)$$

$$M_{\underline{X}}(t) = \left(\sum_{i=1}^3 p_i e^{t_i}\right)^2 = (p_1 e^{t_1} + p_2 e^{t_2} + p_3 e^{t_3})^2$$

PMF of the Hypergeometric Distribution

A random variable X follows the hypergeometric distribution if its probability mass function (pmf) is given by

$$P(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, 2, \dots, K \text{ if } K < n \text{ \& } x = 0, 1, 2, \dots, n \text{ if } n < K$$

where,

- N is the population size,
- K is the number of success in the population at the beginning of the experiment,
- n is the number of draws (i.e. quantity drawn in each trial),
- k is the number of observed successes

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Mean of the Hypergeometric Distribution

$$\text{Mean} = \mu = E(X) = n \frac{K}{N}$$

Variance of the Hypergeometric Distribution

$$\text{Variance} = \sigma^2 = n \frac{K}{N} \frac{(N-K)}{N} \frac{N-n}{N-1}$$

Short question March 2022

PMF of the Poisson Distribution

$$p(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & x = 0, 1, 2, \dots, \lambda > 0 \\ 0 & \text{elsewhere,} \end{cases}$$

Mean and Variance of the Poisson Distribution

$$\text{Mean} = \text{Variance} = \lambda > 0$$

$$\mu = \sigma^2 = \lambda$$

Coefficient of Variation of the Poisson Distribution

$$COV = \frac{\sigma}{\mu} = \frac{\sigma}{\mu} \times 100\% = \frac{\sqrt{\lambda}}{\lambda} \times 100\% = \frac{1}{\sqrt{\lambda}} \times 100\%$$

MGF of the Poisson Distribution

$$M(t) = \sum_x e^{tx} p(x) = \sum_{x=0}^{\infty} e^{tx} \frac{m^x e^{-m}}{x!} = e^{-m} \sum_{x=0}^{\infty} \frac{(me^t)^x}{x!} = e^{-m} e^{me^t} = e^{m(e^t - 1)}$$

PDF of the Continuous Uniform Distribution

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$$

CDF of the Continuous Uniform Distribution

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases}$$

Mean of the Continuous Uniform distribution

$$\mu = E(X) = \frac{(a+b)}{2}$$

Variance of the Continuous Uniform distribution

$$\sigma^2 = \text{Var}(X) = \frac{(b-a)^2}{12}$$

PDF of the Exponential Distribution

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0 \end{cases}$$

Let $\theta = \frac{1}{\lambda}$ then, mean = θ , and

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}} & x \geq 0, \theta > 0 \\ 0 & x < 0 \end{cases}$$

CDF of the Exponential Distribution

$$F(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0 \end{cases}$$

or

$$F(x; \theta) = \begin{cases} 1 - e^{-\frac{x}{\theta}} & x \geq 0, \theta > 0 \\ 0 & x < 0 \end{cases}$$

Mean of the Exponential Distribution

$$\mu = E(X) = \frac{1}{\lambda}$$

Variance and Standard Deviation of the Exponential Distribution

$$\text{Var}(X) = \sigma^2 = \frac{1}{\lambda^2}$$

$$S.D(X) = \sigma = \frac{1}{\lambda}$$

Coefficient of Variation of the Exponential Distribution

$$CV = \frac{\sigma}{\mu} = \frac{1/\lambda}{1/\lambda} = \frac{\lambda}{\lambda} = 1$$

MGF of the Exponential Distribution

$$M_0(t) = E[e^{tX}] = \frac{\lambda}{\lambda - t}$$

Mcq March 2022

PDF of the Gamma Distribution

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & 0 < x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Gamma Function

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy$$

CDF of the Gamma Distribution

$$F(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} \gamma\left(\alpha, \frac{x}{\beta}\right) & \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where this $\gamma\left(\alpha, \frac{x}{\beta}\right)$ is the incomplete gamma function.

Incomplete Gamma Function

In mathematics, the **upper** and **lower incomplete gamma functions** are types of special functions which arise as solutions to various mathematical problems such as certain integrals.

The upper incomplete gamma function is defined as: $\Gamma(\alpha, y) = \int_y^{\infty} t^{\alpha-1} e^{-t} dt$

whereas the lower incomplete gamma function is defined as: $\gamma(\alpha, y) = \int_0^y t^{\alpha-1} e^{-t} dt$

It is interesting to note that $\gamma(\alpha, y) + \Gamma(\alpha, y) = \Gamma(\alpha)$.

Mean of the Gamma Distribution

$$\mu_x = E(X) = \alpha\beta$$

Variance of the Gamma Distribution

$$\text{Var}(X) = \sigma_x^2 = \alpha\beta^2$$

Mode of the Gamma Distribution

$$\text{Mode} = (\alpha - 1)\beta$$

MGF of the Gamma Distribution

$$M(t) = (1 - \beta t)^{-\alpha} \text{ for } t < \frac{1}{\beta}$$

PDF of the Beta Distribution

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 \leq x \leq 1; \alpha, \beta > 0$$

$$\text{where, } B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

CDF of the Beta Distribution

The formula for the cumulative distribution function of the beta distribution is also called the **incomplete beta function ratio** (commonly denoted by I_x) and is defined as

$$F(x) = I_x(\alpha, \beta) = \frac{\int_0^x t^{\alpha-1} (1-t)^{\beta-1} dt}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1; \alpha, \beta > 0$$

Mean of the Beta Distribution

$$\mu = E(X) = \frac{\alpha}{(\alpha + \beta)}$$

Variance of the Beta Distribution

$$\text{Var}(X) = \sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

Mode of the Beta Distribution

$$\text{Mode} = \frac{\alpha - 1}{\alpha + \beta - 2}$$

PDF of the Normal Distribution

A random variable X is said to have a normal distribution if its pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty.$$

μ is the mean or expectation, $\sigma > 0$ is the standard deviation, and σ^2 is the variance.

CDF of the Normal Distribution

$$F(x) = \frac{1}{2} \left[1 + \text{erf} \left(\frac{x - \mu}{\sigma\sqrt{2}} \right) \right] \quad \text{for } -\infty < x < \infty$$

$\text{erf}(z)$ is the “error function” defined by $\text{erf}(z) \equiv \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$

Mean of the Normal Distribution

$$\text{Mean} = E(X) = \mu$$

Median of the Normal Distribution

$$\text{Median} = \mu$$

Mode of the Normal Distribution

$$\text{Mode} = \mu$$

Mean of the Normal Distribution

$$\text{Var}(X) = \sigma^2$$

Mean of the Normal Distribution

$$M(t) = E(e^{tX}) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}, \quad \text{for } -\infty < t < \infty$$

Cumulant Generating Function of the Normal distribution

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$$K(t) = \ln E(e^{tX}) = \ln e^{\mu t + \frac{1}{2}\sigma^2 t^2} = \mu t + \frac{\sigma^2 t^2}{2} \cdot \ln e = \mu t + \frac{\sigma^2 t^2}{2} (1) = \mu t + \frac{\sigma^2 t^2}{2}$$

The cumulants of the normal distribution with expected value μ and variance σ^2 are

$$K_1 = \mu, \quad K_2 = \sigma^2$$

and

$$K_3 = K_4 = \dots = 0.$$

Standard Normal Distribution

The standard normal distribution is a **special** case of the normal distribution.

It is the distribution that occurs when a normal random variable has a **mean** of zero and a **standard deviation** of one.

PDF of the Standard Normal Distribution

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{z^2}{2}\right)}, \quad -\infty < z < \infty$$

Mean of the Standard Normal Distribution

$$\mu = E(Z) = \int_{-\infty}^{\infty} z f_Z(z) dz = 0$$

Variance of the Standard Normal Distribution

$$\text{Var}(Z) = \sigma^2 = 1$$

Semi-Inter Quartile Range or Quartile Deviation

$$Q.D = \frac{Q_3 - Q_1}{2}$$

Some Relationships for Standard Normal Distribution

- $Q / \sigma = 0.6745$ or $Q = 0.6745\sigma$
- $Q_1 = \mu - 0.6745\sigma$ and $Q_3 = \mu + 0.6745\sigma$
- $x = \mu + \sigma$ and $x = \mu - \sigma$

The normal curve has Points of Inflection which are equidistant from the mean.

Two points of inflection of normal curve are

$$\left[\mu - \sigma, \frac{1}{\sigma\sqrt{2\pi e}} \right] \text{ and } \left[\mu + \sigma, \frac{1}{\sigma\sqrt{2\pi e}} \right].$$

In other words, the points of inflection occur on the right and on the left of the mean at a distance equal to standard deviation and thus the graph of the normal curve are bell-shaped.

Theorem:

For the normal distribution, the **odd** order moments about the mean are all zero.

$$\mu_1 = \mu_3 = \mu_5 = \dots = 0$$

For the normal distribution, the **odd** order moments about the mean are given by

$$\mu_{2n} = (2n-1)(2n-3)\dots 5.3.1 \sigma^{2n}$$

PDF of the Bivariate Normal Distribution

A pair of random variables X_1 and X_2 has a bivariate normal distribution and they are referred to as **jointly normally distributed** random variables **if and only if** their joint probability density is given by

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{\left\{ -\frac{1}{2(1-\rho^2)} \left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho \left(\frac{x_1-\mu_1}{\sigma_1} \right) \left(\frac{x_2-\mu_2}{\sigma_2} \right) + \frac{(x_2-\mu_2)^2}{\sigma_2^2} \right) \right\}}$$

for $-\infty < x_1 < \infty$ and $-\infty < x_2 < \infty$,

where $\mu_1 \in \mathbb{R}, \mu_2 \in \mathbb{R}, \sigma_1 > 0, \sigma_2 > 0$, and $-1 \leq \rho \leq 1$.

This is referred to as the $N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$ distribution.

MGF of the Bivariate Normal Distribution

$$M_{\underline{X}}(t) = M_{\underline{X}}(t_1, t_2) = E \left(e^{\sum_{i=1}^2 t_i X_i} \right)$$

$$M_{\underline{X}}(t) = e^{\mu_1 t_1 + \mu_2 t_2 + \frac{1}{2}(\sigma_1^2 t_1^2 + 2\rho\sigma_1\sigma_2 t_1 t_2 + \sigma_2^2 t_2^2)}, \quad -\infty < t_1 < \infty; -\infty < t_2 < \infty$$

Marginal of the Bivariate Normal Distribution

For any bivariate distribution $f(x, y)$, the marginal distribution of X is given by

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad -\infty < x < \infty$$

$$f_X(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2} - \frac{2\rho(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y}\right]} dy$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_x} e^{-\frac{1}{2}\left(\frac{x-\mu_x}{\sigma_x}\right)^2}, \quad -\infty < x < \infty$$

i.e. the normal distribution with μ_x and Variance σ_x^2 .

$$\text{i.e. } X \sim N(\mu_x, \sigma_x^2) \text{ with } E(X) = \mu_x, \text{ Var}(X) = \sigma_x^2,$$

Similarly, $Y \sim N(\mu_y, \sigma_y^2)$ with $E(Y) = \mu_y$, $\text{Var}(Y) = \sigma_y^2$.

Theorem:

If X and Y have a bivariate normal distribution with correlation coefficient ρ . Then X and Y are independent **if and only if** $\rho = 0$.

Definition of Independence:

Two random variables X and Y are independent, statistically independent or stochastically independent if

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

i.e. if the joint density $f_{XY}(x, y)$ is equal to the product of the marginal density $f_X(x)$ and the marginal density $f_Y(y)$.

Conditional distributions in the case of Bivariate Normal Distribution

The conditional distribution of Y given x can be expressed as follows:

$$N\left(\mu_y + \rho \frac{\sigma_y}{\sigma_x}(x - \mu_x), \sigma_y^2(1 - \rho^2)\right) = N(\tilde{\mu}_y, \tilde{\sigma}_y^2)$$

Always remember in prayers.

MaLiK

discrete uniform distribution:-

PMF

$$P(x) = \begin{cases} \frac{1}{n} & x=1,2,\dots,n \\ 0 & \text{elsewhere} \end{cases}$$

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Mean

$$\mu = \frac{n+1}{2}$$

Variance

$$\frac{n^2-1}{12} = \text{Var}(x)$$

MGF

$$M(t) = \frac{e^t - e^{(n+1)t}}{n(1-e^t)}$$

Write pdf of binomial Distribution March 2022 time.

2.30

Binomial Distribution:-

PMF

$$P(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x=0,1,2,\dots,n \\ 0 & \text{elsewhere} \end{cases}$$

Mean

$$\mu = np$$

Variance

$$\text{Var}(x) = np(1-p)$$

MGF

$$M(t) = [(1-p) + pe^t]^n$$

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real value of t .

Parameters

n and p

Negative Binomial distribution:-

PMF

$$P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$$

$$x = r, r+1, \dots$$

Mean

$$\mu = \frac{r(1-p)}{p}$$

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Variance

$$\sigma^2 = \frac{r(1-p)}{p^2}$$

MGF

$$M(t) = \left(\frac{p}{1-pe^t} \right)^r \quad \text{for } t < -\log p$$

Parameters

r and p .

Geometric distribution:-

PMF

$$P_X(x) = (1-p)^{x-1} p, \quad x=1, 2, \dots$$

Mean

$$\mu = \frac{1}{p}$$

Variance

$$\sigma^2 = \frac{1-p}{p^2}$$

Parameter

p only.

Multinomial distribution:-

PMF

$$f(x_1, \dots, x_k; n, p_1, \dots, p_k) = \begin{cases} \frac{n!}{x_1! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}, & \sum_{i=1}^k x_i = n \\ 0, & \text{elsewhere} \end{cases}$$

Find mean and Variance short question

March 2022

Mean

$$E(X_i) = np_i \quad \text{Obj March 2022}$$

Variance

$$\text{Var}(X_i) = np_i (1 - p_i)$$

Covariance

$$\text{Cov}(X_i, X_j) = -np_i \cdot p_j, \quad i \neq j$$

MGF

$$\begin{aligned} M_{\underline{X}}(t) &= M_{\underline{X}}(t_1, t_2, \dots, t_k) = E\left(e^{\sum_{i=1}^k t_i X_i}\right) \\ &= \left(\sum_{i=1}^k p_i e^{t_i}\right)^n \end{aligned}$$

CGF

$$\begin{aligned} K_{\underline{X}}(t) &= \log(M_{\underline{X}}(t_1, t_2, \dots, t_k)) \\ &= n \log\left(\sum_{i=1}^k p_i e^{t_i}\right) \end{aligned}$$

Hypogeometric distribution:-

PMF Subj March 2022

$$P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

$$x = 0, 1, 2, \dots, K \quad \text{if } K < n$$

$$x = 0, 1, 2, \dots, n \quad \text{if } n < K$$

Subj March 2022

Mean

$$E(X) = n \frac{K}{N} \quad \text{or} \quad E(X) = np$$

$$\text{where } p = \frac{K}{N}$$

Subj March 2022

Variance

$$\text{Var}(X) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$$

or

$$\text{Var}(X) = npq \frac{N-n}{N-1}$$

$$\text{where } p = \frac{K}{N}, q = \frac{N-K}{N}$$

FPC

$$\text{Finite population correlation} = \sqrt{\frac{N-n}{N-1}}$$

Parameters

x, N, n and K .

Poisson distribution:-

PMF

$$P(X) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & x=0,1,\dots, \lambda > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Mean

=

$$\mu = \lambda$$

Variance

=

$$\text{Var}(X) = \lambda$$

Same

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MGF

$$M(t) = e^{m(e^t - 1)}$$

for real values of t .

COV

$$\text{COV} = \frac{\sigma}{\mu} \times 100\%$$

$$= \frac{1}{\sqrt{\lambda}} \times 100\%$$

Obj March 2022

Parameter

=

λ only.

Continuous uniform distribution:-

OR

Rectangular distribution:-

$$\underline{\underline{\text{PDF}}}$$
$$f(x) = \begin{cases} \frac{1}{b-a} & , \quad a \leq x \leq b \\ 0 & , \quad x < a, x > b. \end{cases}$$

$$\underline{\underline{\text{CDF}}}$$
$$F(x) = \begin{cases} 0 & , \quad x < a \\ \frac{x-a}{b-a} & , \quad a \leq x \leq b \\ 1 & , \quad x > b \end{cases}$$

Mean Obj March 2022

$$E(x) = \frac{a+b}{2}$$

Var(x)

$$\text{Var}(x) = \frac{(b-a)^2}{12}$$

Exponential distribution:-

PDF

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0 \end{cases}$$

$$f(x, \theta) = \begin{cases} \frac{1}{\theta} e^{-x/\theta} & x \geq 0, \theta > 0 \\ 0 & x < 0 \end{cases}$$

CDF

$$F(x, \lambda) = \begin{cases} 1 - e^{-\lambda x} & , \quad x \geq 0, \lambda > 0 \\ 0 & , \quad x < 0 \end{cases}$$

$$F(x, \theta) = \begin{cases} 1 - e^{-x/\theta} & , \quad x \geq 0, \theta > 0 \\ 0 & , \quad x < 0 \end{cases}$$

Mean

$$\mu = \frac{1}{\lambda} = \text{Standard dev.}$$

Variance

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

Cov

$$\text{Cov} = 1$$

MGF (exponential distribution)

$$M_0(t) = \frac{\lambda}{\lambda - t}, \text{ for } t < \lambda$$

Gamma distribution:-

PDF

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, & 0 < x < \infty \\ & \alpha, \beta > 0 \end{cases}$$

cluster

CDF

$$\begin{cases} \frac{1}{\Gamma(\alpha)} \gamma\left(\alpha, \frac{x}{\beta}\right), & \alpha, \beta > 0 \\ 0 & \text{cluster} \end{cases}$$

Mean

$$\mu_x = \alpha\beta$$

Variance

$$\sigma_x^2 = \alpha\beta^2$$

MGF

$$M(t) = (1 - \beta t)^{-\alpha} \text{ for } t < \frac{1}{\beta}$$

Beta distribution:-

PDF

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$0 \leq x \leq 1$$

$$\alpha, \beta > 0$$

CDF

$$F(x) = I_x(\alpha, \beta) = \frac{\int_0^x t^{\alpha-1} (1-t)^{\beta-1} dt}{B(\alpha, \beta)}$$

$$0 \leq x \leq 1, \alpha, \beta > 0.$$

Mean

$$\mu = \frac{\alpha}{\alpha + \beta}$$

Variance

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

Mode

$$\text{mode} = \frac{\alpha - 1}{\alpha + \beta - 2}$$